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- (54) **BACK-SIDE POWER DELIVERY WITH GLASS SUPPORT AT THE FRONT**

- (71) Applicant: **Intel Corporation**, Santa Clara, CA
(US)

- (72) Inventors: **Abhishek A. Sharma**, Hillsboro, OR (US); **Wlfred Gomes**, Portland, OR (US); **Telesphor Kamgaing**, Chandler, AZ (US)

- (73) Assignee: **Intel Corporation**, Santa Clara, CA
(US)

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CPC ***H01L 23/5286*** (2013.01); ***H01L 21/768***
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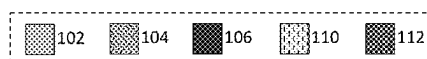
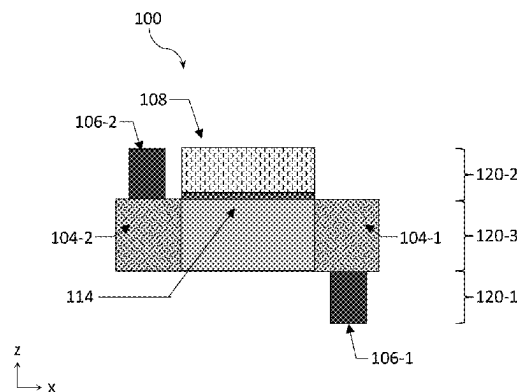
Assistant Examiner — Sophia W Kao

(74) *Attorney, Agent, or Firm* — Akona IP PC

- (57) **ABSTRACT**

Embodiments of the present disclosure are based on using transistors with back-side contacts. Such transistors enable back-side power delivery to IC components (e.g., transistors, etc.) of an IC structure, which may be more advantageous than front-side power delivery in some implementations. Embodiments of the present disclosure are further based on recognition that using a glass support structure at the front side of an IC structure with back-side power delivery may advantageously reduce parasitic effects in the IC structure, e.g., compared to using a silicon-based support structure at the front.

20 Claims, 13 Drawing Sheets



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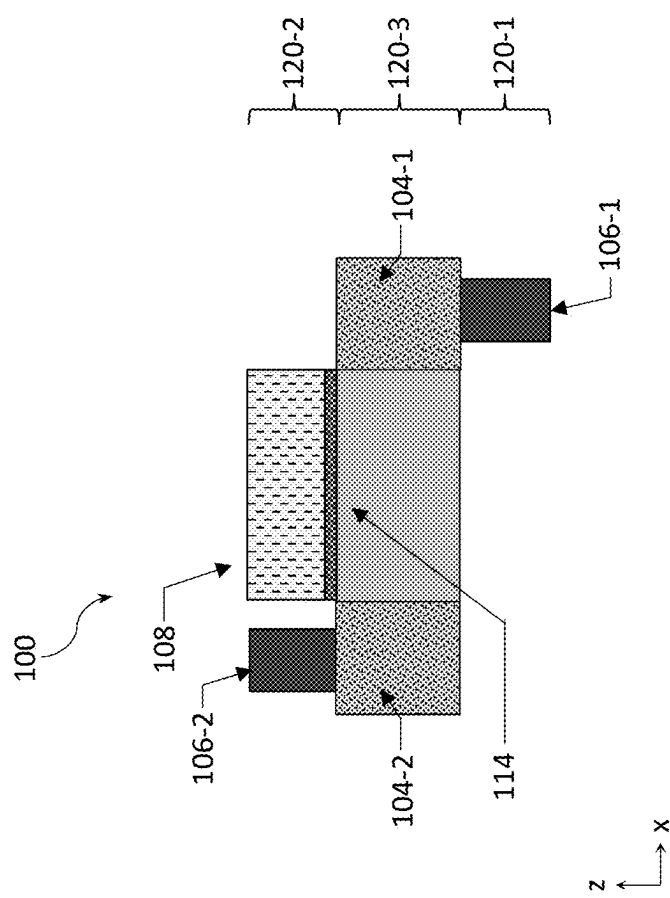
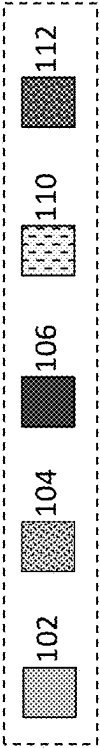


FIG. 1



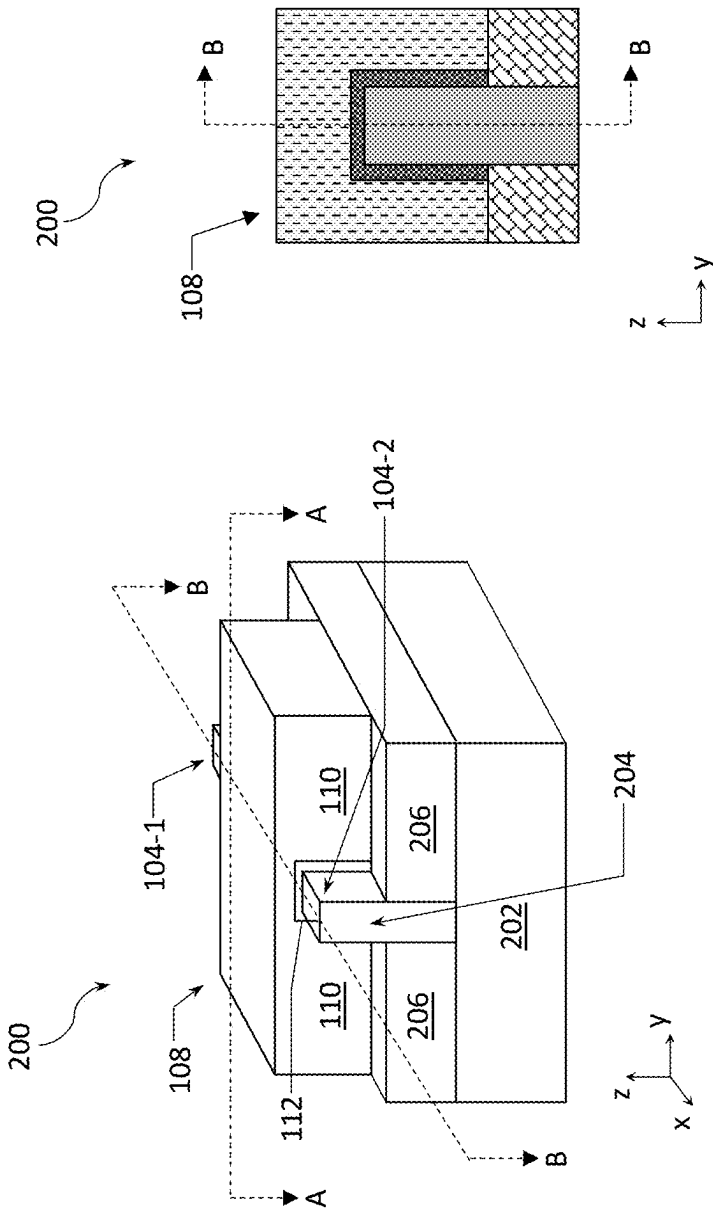


FIG. 2A

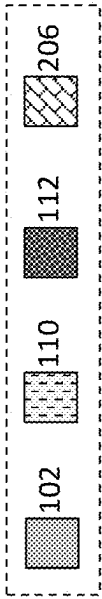
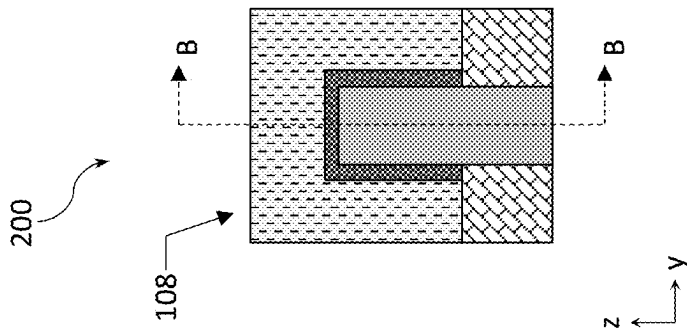


FIG. 2B



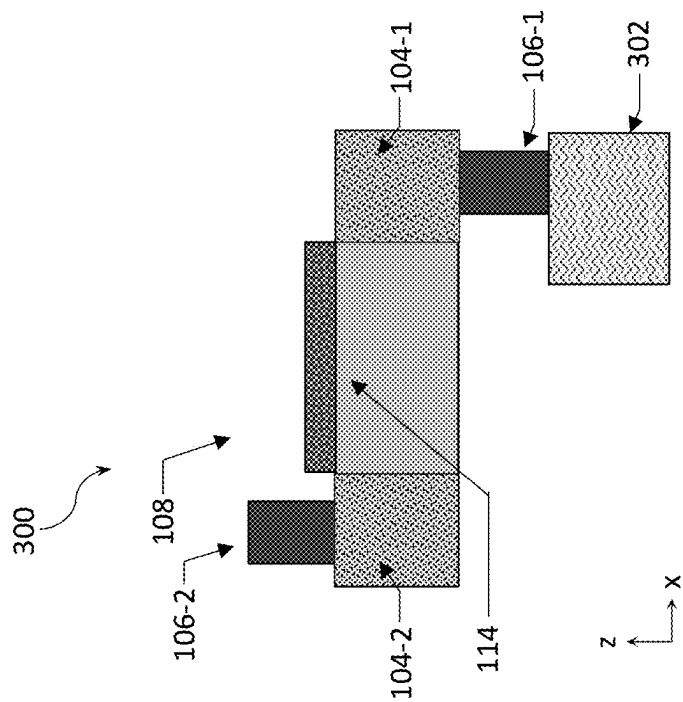
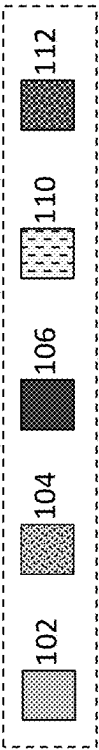


FIG. 3



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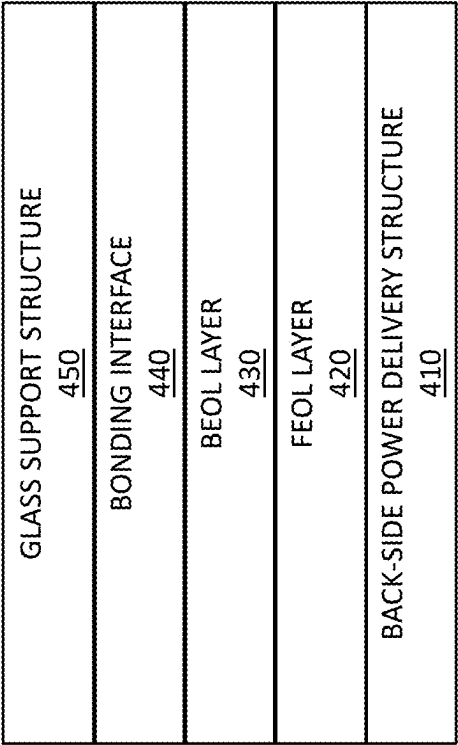


FIG. 4

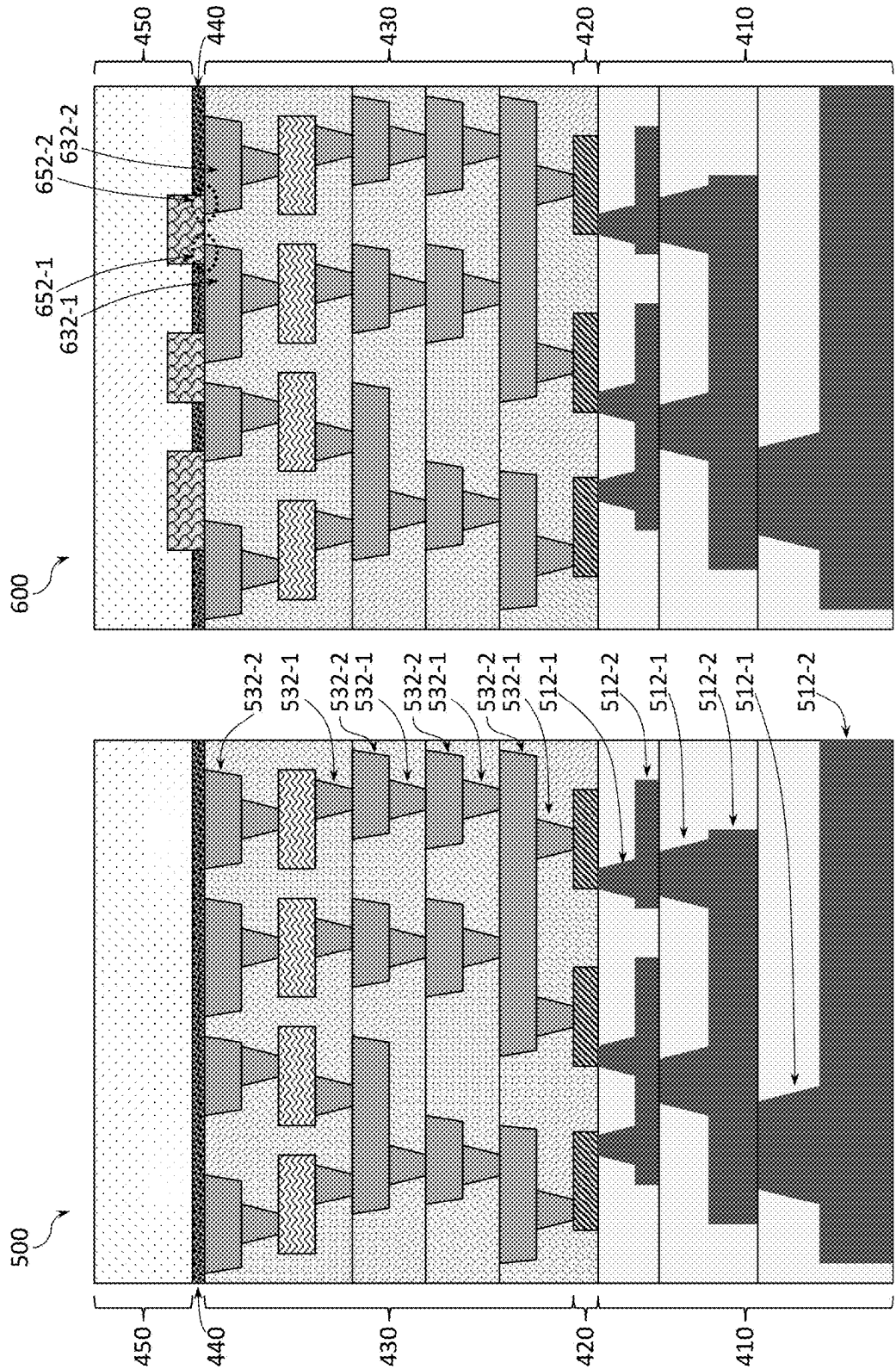
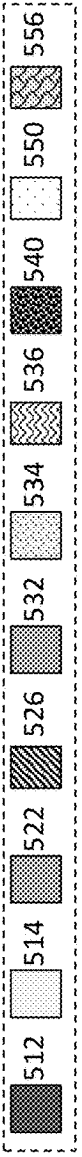


FIG. 5

FIG. 6



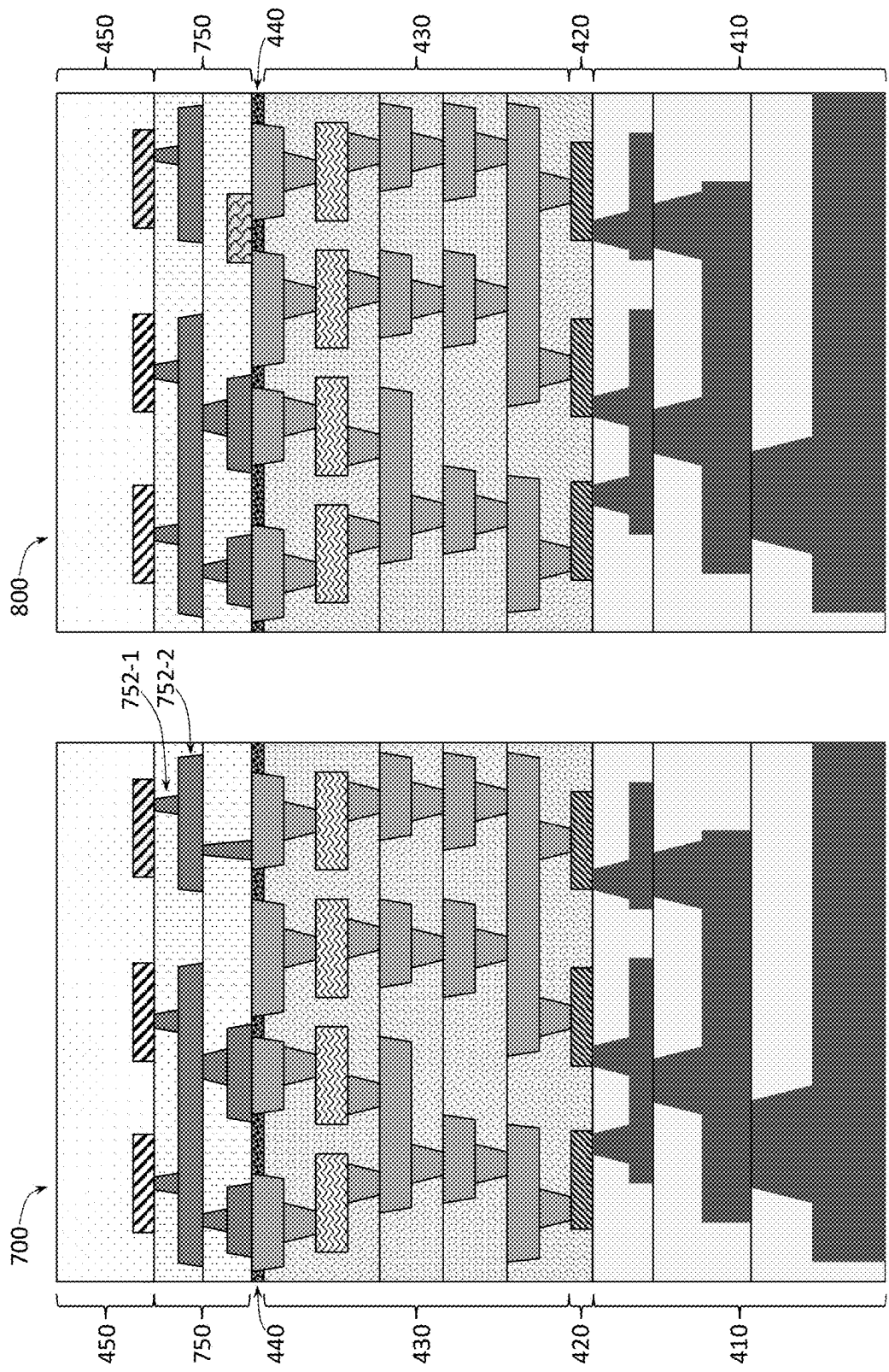
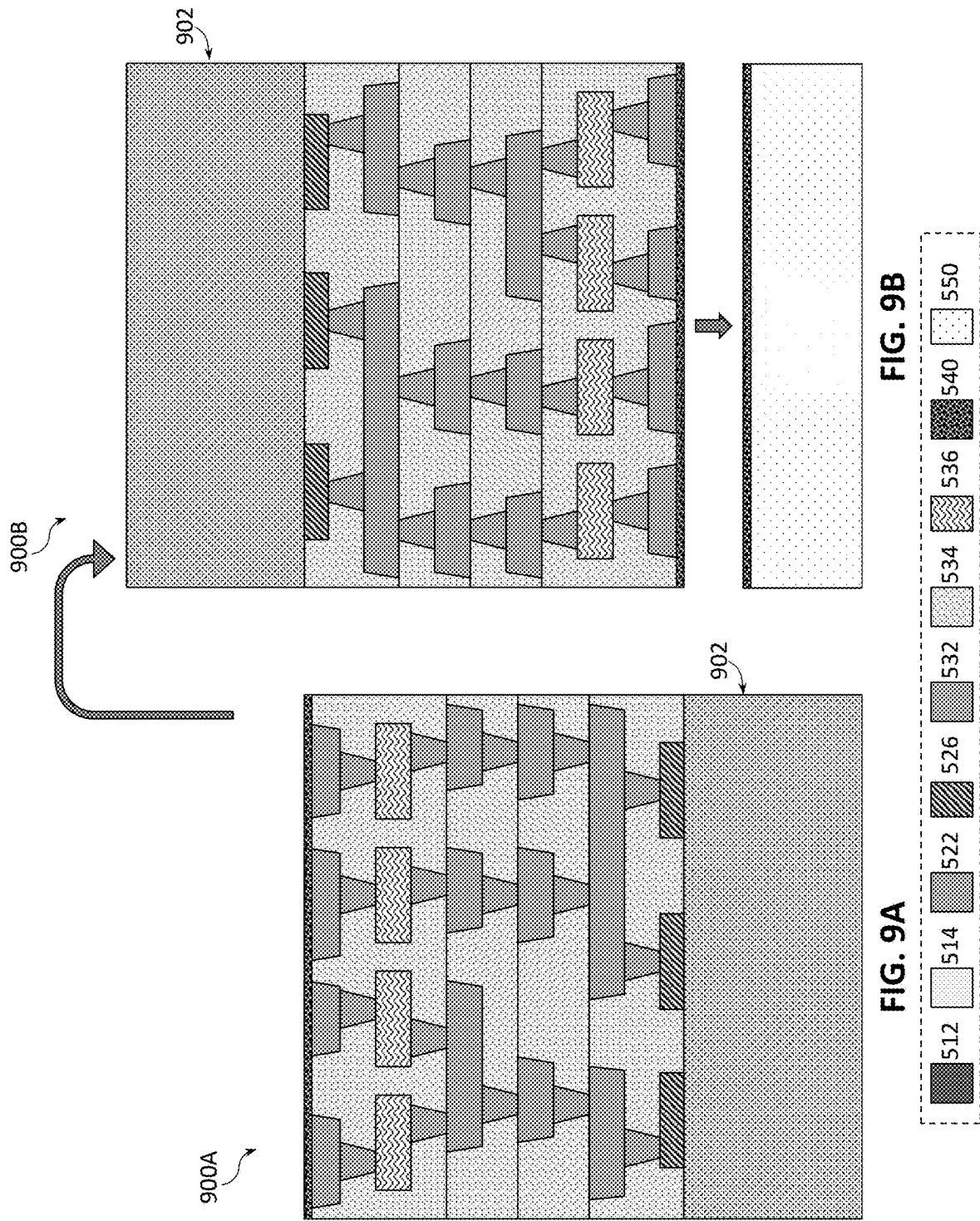
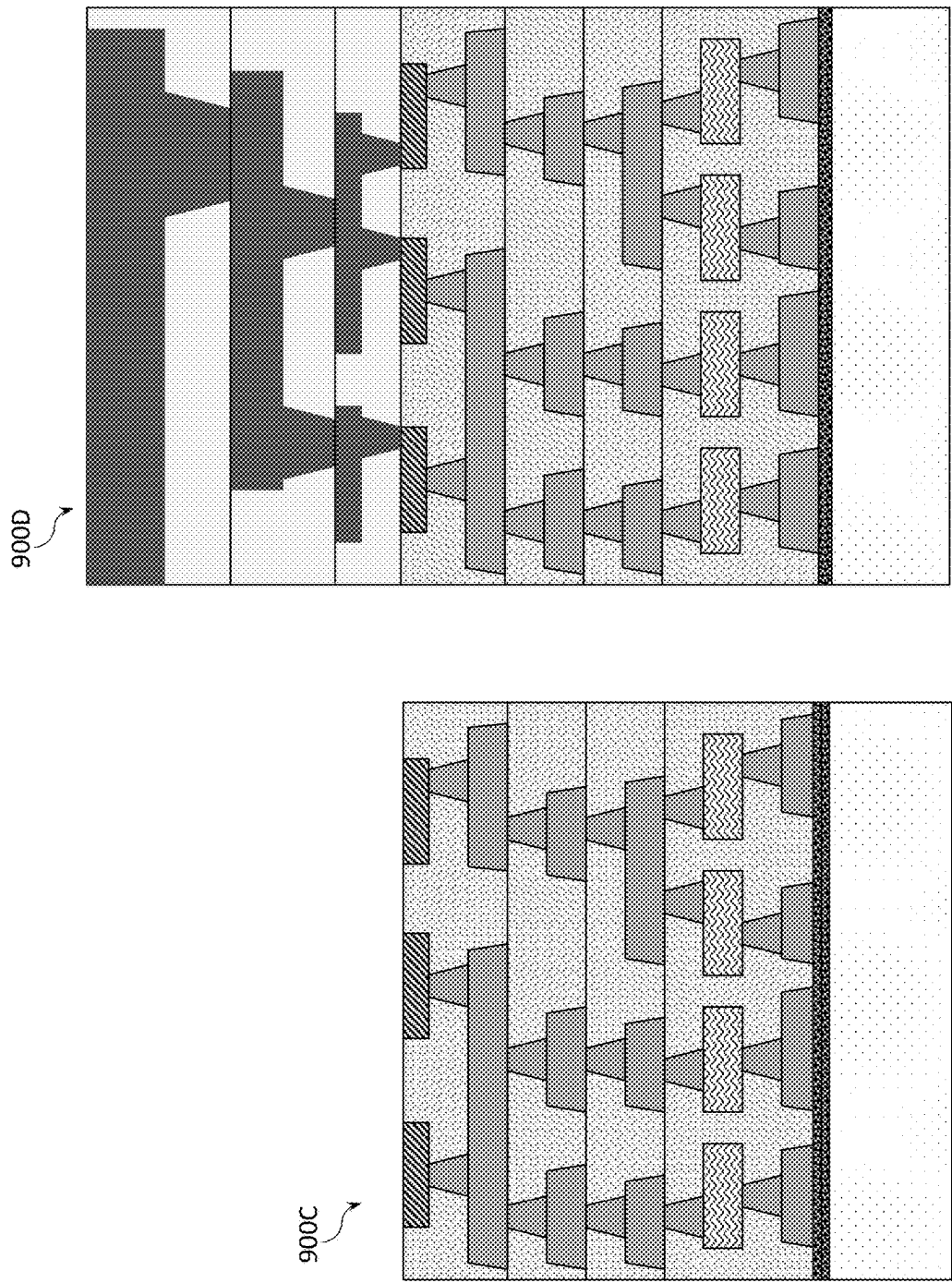
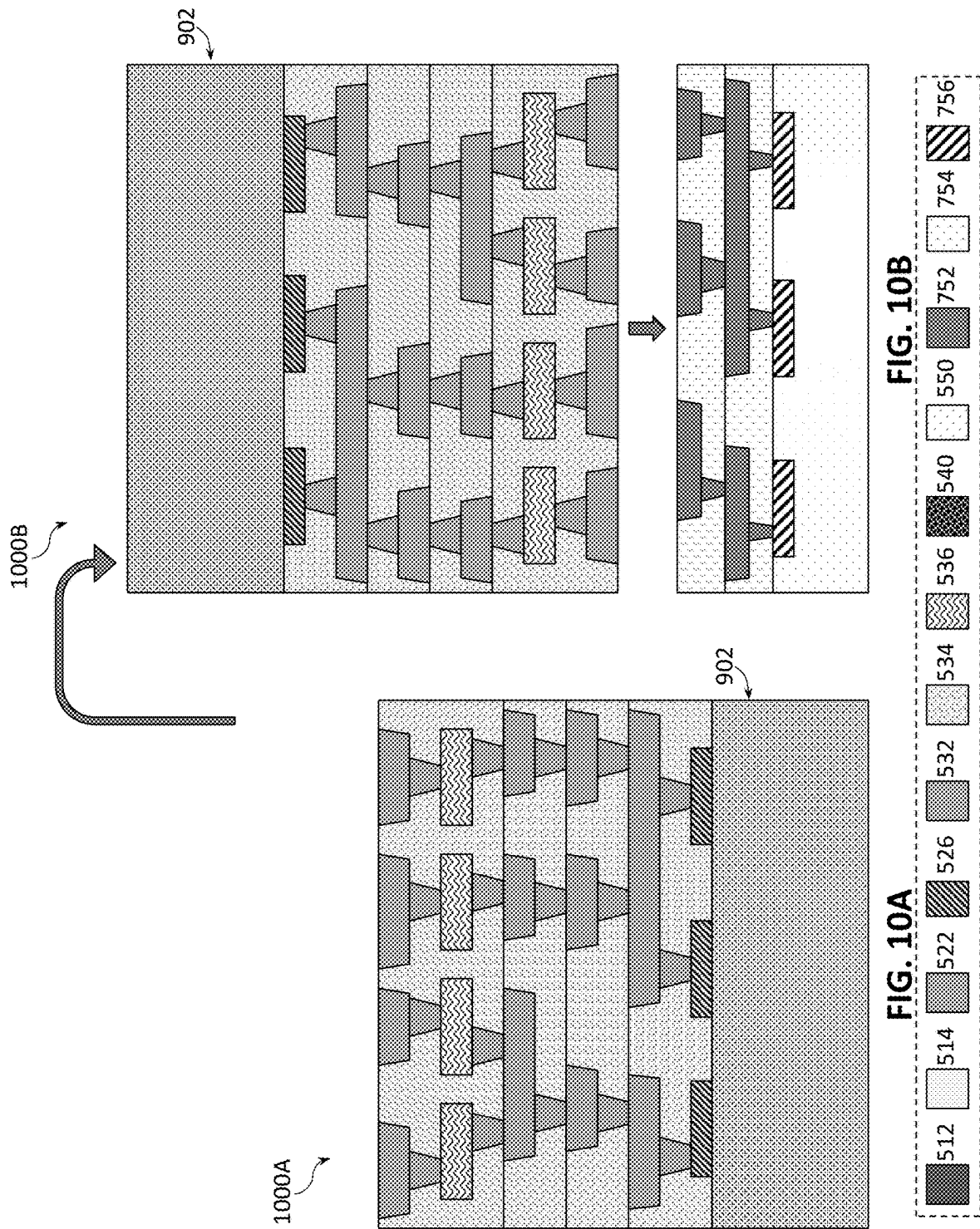


FIG. 7

FIG. 8







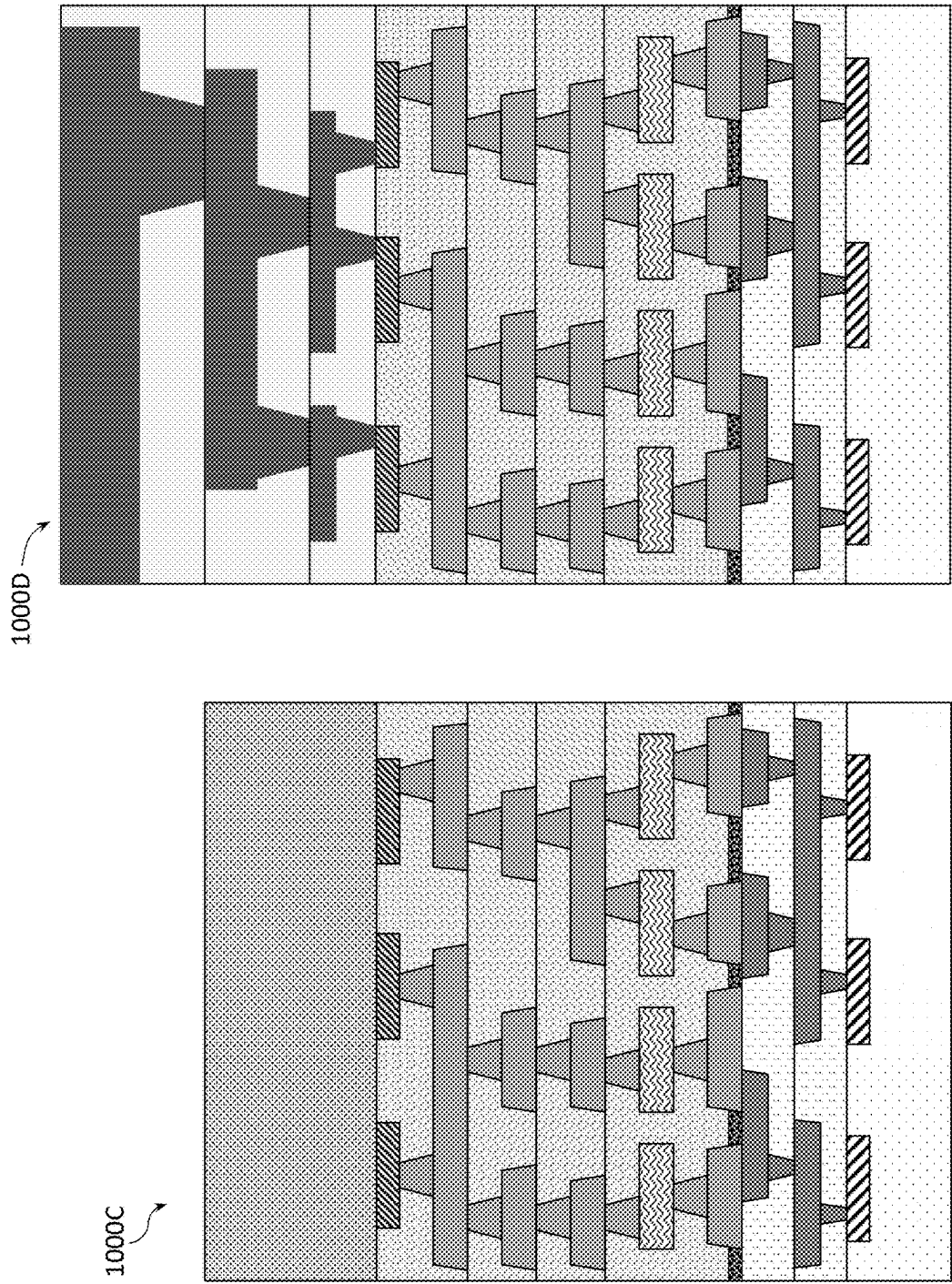
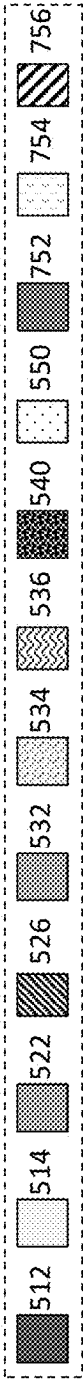


FIG. 10C

FIG. 10D



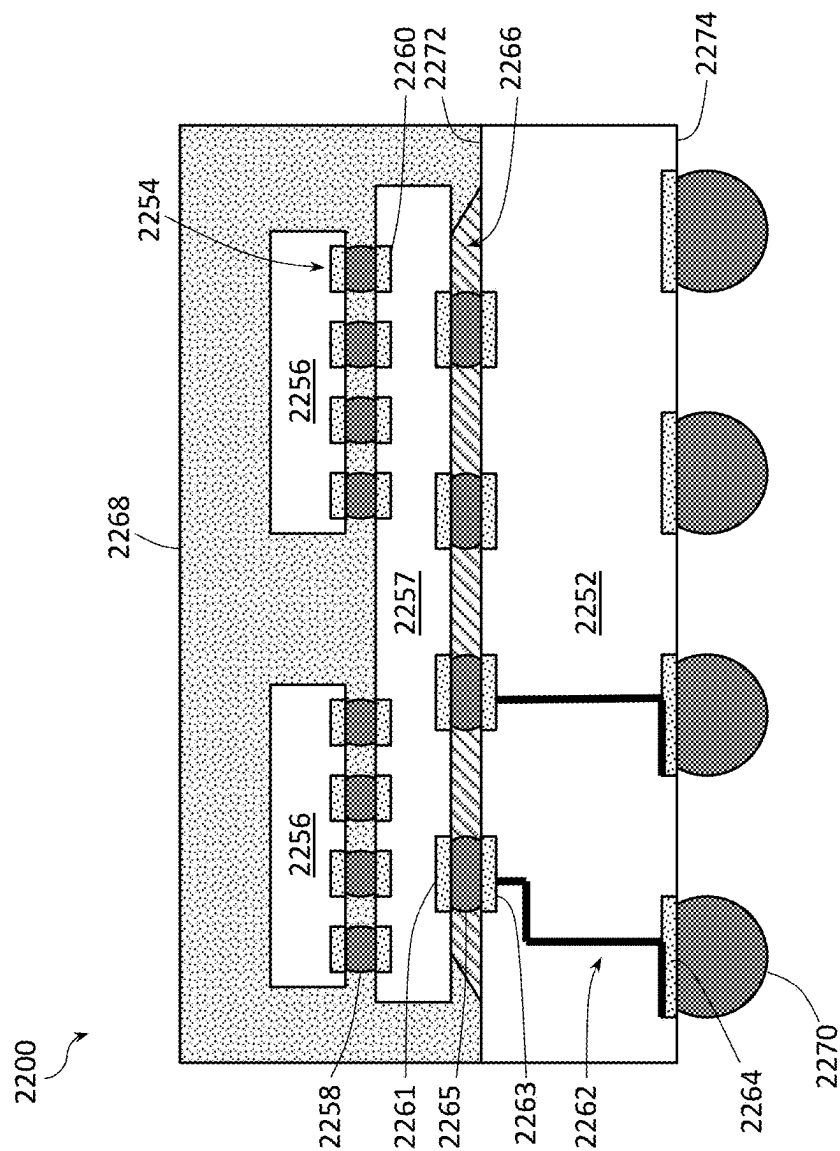


FIG. 11

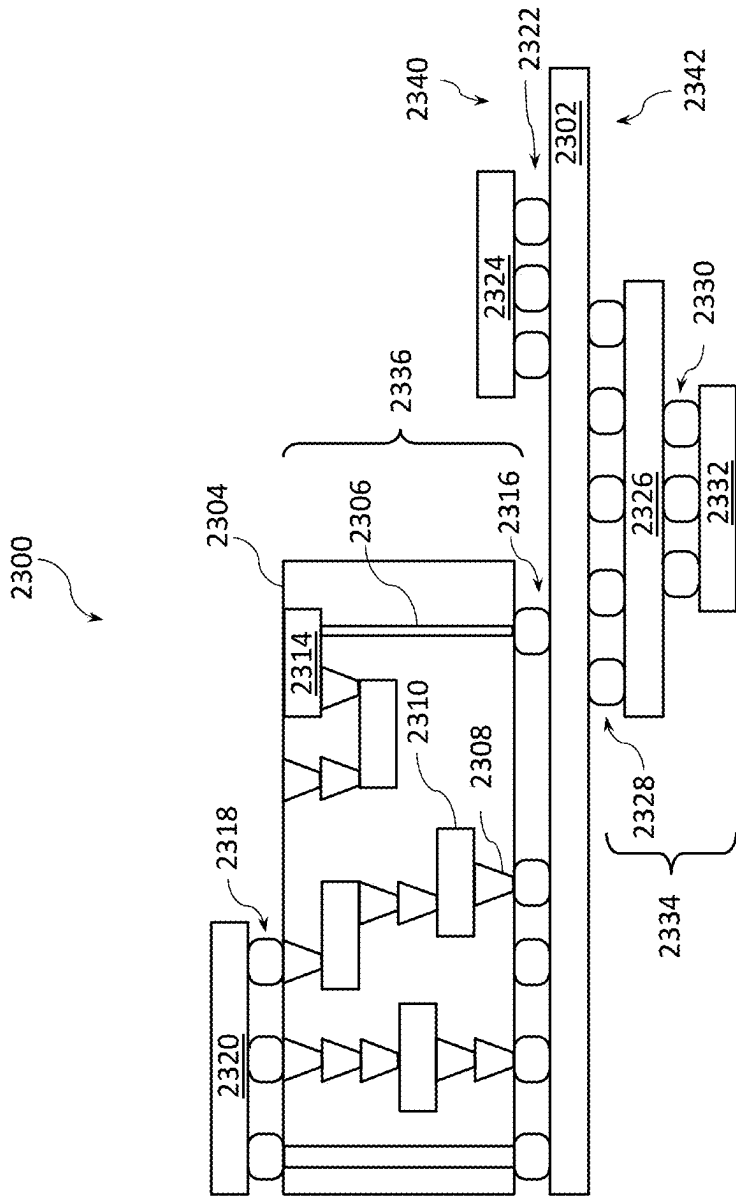


FIG. 12

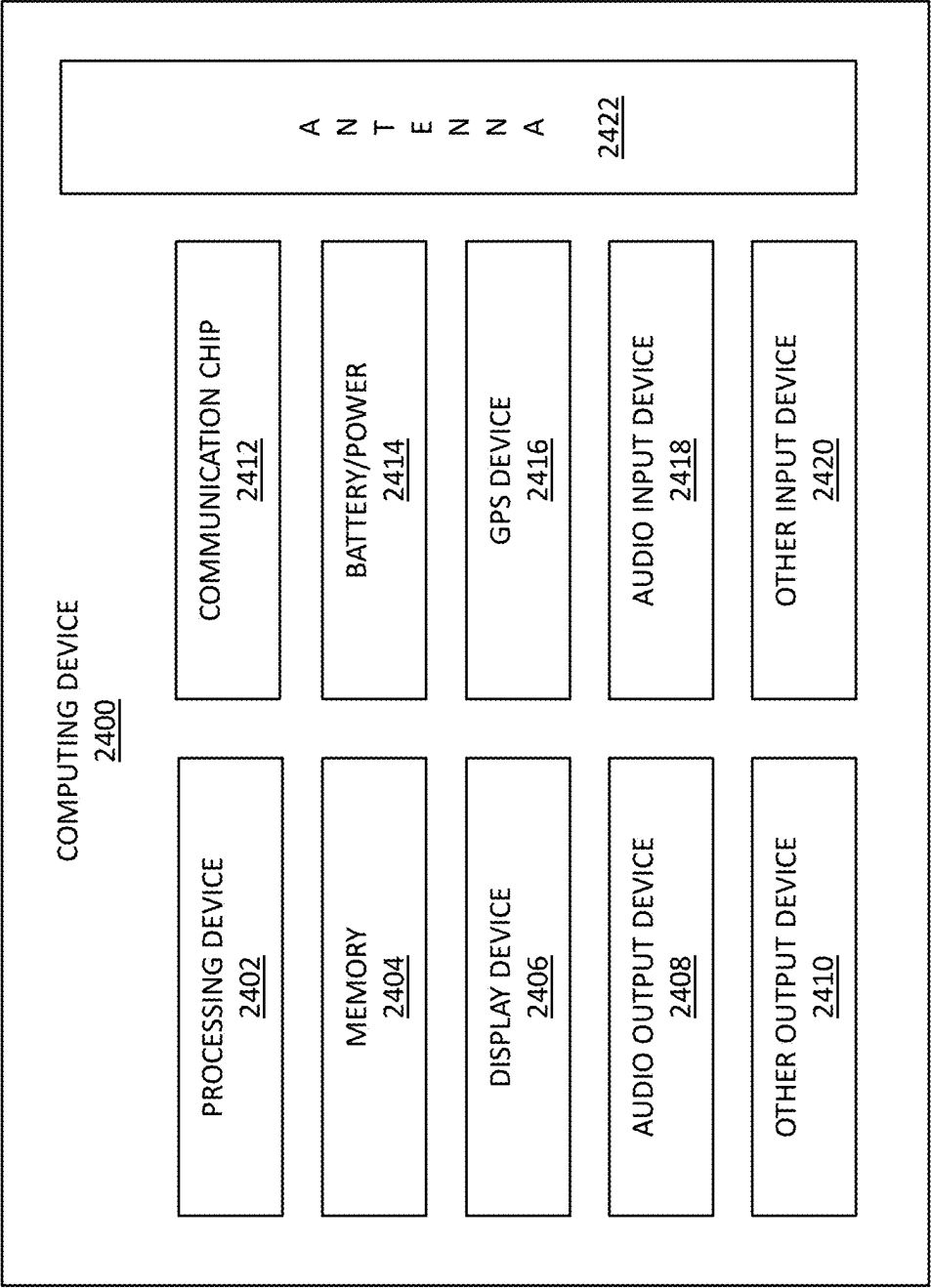


FIG. 13

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BACK-SIDE POWER DELIVERY WITH GLASS SUPPORT AT THE FRONT

BACKGROUND

For the past several decades, the scaling of features in integrated circuits has been a driving force behind an ever-growing semiconductor industry. Scaling to smaller and smaller features enables increased densities of functional units on the limited real-estate of semiconductor chips. For example, shrinking transistor size allows for the incorporation of an increased number of memory or logic devices on a chip, lending to the fabrication of products with increased capacity. The drive for the ever-increasing capacity, however, is not without issue. The necessity to optimize the performance of each device and each interconnect becomes increasingly significant.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

FIG. 1 provides a schematic illustration of a cross-sectional view of an example transistor with a back-side contact according to some embodiments of the present disclosure.

FIGS. 2A-2B are perspective and cross-sectional views, respectively, of an example transistor with a back-side contact implemented as a FinFET, according to some embodiments of the present disclosure.

FIG. 3 provides a schematic illustration of a cross-sectional view of an example memory cell that includes a transistor with a back-side contact, according to some embodiments of the present disclosure.

FIG. 4 provides a block diagram of an integrated circuit (IC) assembly with back-side power delivery and glass support at the front, according to some embodiments of the present disclosure.

FIGS. 5-8 provide schematic illustrations of an IC assembly with back-side power delivery and glass support at the front, according to various embodiments of the present disclosure.

FIGS. 9A-9D illustrate a first example method of forming an IC assembly with back-side power delivery and glass support at the front, according to some embodiments of the present disclosure.

FIGS. 10A-10D illustrate a second example method of forming an IC assembly with back-side power delivery and glass support at the front, according to some embodiments of the present disclosure.

FIG. 11 is a cross-sectional side view of an IC package that may include an IC assembly with back-side power delivery and glass support at the front in accordance with any of the embodiments disclosed herein.

FIG. 12 is a cross-sectional side view of an IC device assembly that may include an IC assembly with back-side power delivery and glass support at the front in accordance with any of the embodiments disclosed herein.

FIG. 13 is a block diagram of an example computing device that may include an IC assembly with back-side

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power delivery and glass support at the front in accordance with any of the embodiments disclosed herein.

DETAILED DESCRIPTION

Overview

The systems, methods and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for all of the desirable attributes disclosed herein. Details of one or more implementations of the subject matter described in this specification are set forth in the description below and the accompanying drawings.

For purposes of illustrating IC assemblies with back-side power delivery and glass support at the front as described herein, it might be useful to first understand phenomena that may come into play in certain IC arrangements. The following foundational information may be viewed as a basis from which the present disclosure may be properly explained. Such information is offered for purposes of explanation only and, accordingly, should not be construed in any way to limit the broad scope of the present disclosure and its potential applications.

Monolithic ICs generally include a number of transistors, such as metal oxide semiconductor (MOS) field-effect transistors (FETs) (MOSFETs), fabricated over a planar substrate, such as a silicon wafer. While Moore's Law has held true for decades within the IC industry, lateral scaling of IC dimensions is becoming more difficult with MOSFET gate dimensions now below 20 nanometers. As device sizes continue to decrease, there will come a point where it becomes impractical to continue standard planar scaling. This inflection point could be due to economics or physics, such as prohibitively high capacitance, or quantum-based variability. Stacking of transistors in a third dimension, typically referred to as vertical scaling, or three-dimensional (3D) integration, is therefore a promising path toward greater transistor density.

While 3D integration may be achieved at a package level, for example by stacking separately manufactured chips, a monolithic 3D approach offers the greatest inter-layer interconnect density, allowing 3D circuits, such as 3D logic circuits, to be constructed at the lowest level and the tightest circuit density. Realizing a monolithic 3D IC architecture with favorable metrics in terms of power, performance, and footprint area is not a trivial task and further improvements are always desirable.

Embodiments of the present disclosure are based on using transistors with back-side contacts. Conventional front end of line (FEOL) transistors have both source and drain contacts on one side of the transistor, usually on the side facing away from the substrate. In contrast to the approaches of building logic and memory devices with such conventional FEOL transistors, various embodiments of the present disclosure provide transistors, various IC devices incorporating such transistors (e.g., logic devices, memory cells and arrays, etc.), as well as associated methods and larger devices, in which a transistor has at least one source or drain (S/D) contact on one side and another S/D contact on the other side. One side of a transistor may be referred to as a "front side" while the other side may be referred to as a "back side," where, in general, in the context of the present disclosure, a "side" of a transistor refers to a region or a layer either above or below a layer of the channel material of the transistor. Thus, transistors described herein may have one of the S/D contacts on the front side (such contacts referred to as "front-side contacts") and the other one of their S/D contacts on the back side (such contacts referred to as

“back-side contacts”). In further embodiments, both S/D contacts of at least some of the transistors used in IC assemblies described herein may be on the back side of the transistor. In the following, transistors having one front-side and one back-side S/D contacts, as well as transistors having two back-side S/D contacts, may be simply referred to as “transistors with back-side contacts.”

Using transistors with back-side contacts provides several advantages and enables unique architectures that were not possible with conventional, FEOL logic transistors with both S/D contacts being on one side. One advantage is that such transistors enable back-side power delivery to IC components (e.g., transistors, etc.) of an IC structure, i.e., delivery of power from the back side of the IC structure. Back-side power delivery may be more advantageous than front-side power delivery in some implementations, e.g., in monolithic 3D IC architectures. Another advantage is that such transistors may be moved to the back end of line (BEOL) layers of an advanced complementary metal oxide semiconductor (CMOS) process. Yet another advantage is that implementing at least some of the transistors with their S/D contacts on different sides allows substantial flexibility to making electrical connections to these transistors. Consequently, at least portions of logic devices and memory cells incorporating such transistors may be provided in different layers above the support structure, thus enabling 3D integration of memory and logic devices and, in particular, enabling a stacked architecture with many layers of memory and/or logic devices. Providing 3D memory and/or logic devices allows significantly increasing density of these devices (e.g., density of memory cells in a memory array) having a given footprint area (the footprint area being defined as an area in a plane of the substrate, or a plane parallel to the plane of the substrate, i.e., the x-y plane of an example coordinate system shown in the drawings of the present disclosure), or, conversely, allows significantly reducing the footprint area of a structure with a given density of memory and/or logic devices.

When back-side power delivery is implemented, besides interconnects for delivering power, a back-side power delivery structure may be include various IC devices (e.g., capacitors, inductors, resistors, etc.) for reducing the parasitic effects of the assembly, e.g., for reducing parasitic effects associated with the interconnects used for power delivery. However, as more and more IC components are implemented on the front side of an IC structure, the density of power interconnects at the backside increases to the point that it becomes challenging to also implement additional IC devices for reducing the parasitic effects of the assembly.

Embodiments of the present disclosure are based on recognition that using a glass support structure at the front side of an IC structure with back-side power delivery may advantageously reduce parasitic effects in the IC structure, e.g., compared to using a silicon-based (Si) support structure at the front. As used herein, the term “glass support structure” refers to any support structure that has a dielectric constant lower than that of Si, e.g., lower than about 11. Such a glass support structure may include any type of glass materials in some embodiments, since glass has dielectric constants in a range between about 5 and 10.5. However, in some embodiments, what is described herein as a glass support structure may include materials other than glass, e.g., mica, as long as those materials have sufficiently low dielectric constants. Arranging a support structure with a dielectric constant lower than that of Si at the front of an IC structure may advantageously decrease various parasitic effects associated with the IC structure, since they are typically proportional to the dielectric constant of the sur-

rounding medium. Furthermore, arranging such a support structure enables implementing at least some of the additional IC devices for reducing the parasitic effects of the assembly at the front of the IC structure, thus advantageously augmenting the back-side power delivery without crowding the valuable real-estate of the back-side power interconnects.

An example IC assembly includes an FEOL layer with a plurality of FEOL devices, a back-side power delivery structure with a plurality of power interconnects electrically coupled to (e.g., in electrically conductive contact with at least portions of) various ones of the plurality of FEOL devices, a BEOL layer with a plurality of BEOL interconnects electrically coupled to (e.g., in electrically conductive contact with at least portions of) one or more of the plurality of FEOL devices, and a glass support structure (e.g., at least a portion of a glass wafer), where the FEOL layer is between the back-side power delivery structure and the BEOL layer, and the BEOL layer is between the FEOL layer and the glass support structure.

In the context of the present disclosure, the term “above” may refer to being further away from the support structure or the FEOL of an IC device, while the term “below” refers to being closer towards the support structure or the FEOL of the IC device.

In the following, some descriptions may refer to a particular side of the transistor being referred to as a front side and the other side being referred to as a back side to illustrate the general concept of transistors having their S/D contacts on different sides. However, unless specified otherwise, which side of a transistor is considered to be a front side and which side is considered to be a back side is not important. Therefore, descriptions of some illustrative embodiments of the front and back sides provided herein are applicable to embodiments where the designation of front and back sides may be reversed, as long as one of the S/D contacts for a transistor is provided on one side and another one—on the other, with respect to the channel layer. Furthermore, some descriptions may refer to a particular S/D region or contact being either a source region/contact or a drain region/contact. However, unless specified otherwise, which region/contact of a transistor is considered to be a source region/contact and which region/contact is considered to be a drain region/contact is not important because, as is common in the field of FETs, designations of source and drain are often interchangeable. Therefore, descriptions of some illustrative embodiments of the source and drain regions/contacts provided herein are applicable to embodiments where the designation of source and drain regions/contacts may be reversed.

While some descriptions provided herein may refer to transistors being top-gated transistors, embodiments of the present disclosure are not limited to only this design and include transistors of various other architectures, or a mixture of different architectures. For example, in various embodiments, transistors with back-side S/D contacts, described herein, may include bottom-gated transistors, top-gated transistors, FinFETs, nanowire transistors, planar transistors, etc., all of which being within the scope of the present disclosure. Furthermore, although descriptions of the present disclosure may refer to logic devices or memory cells provided in a given layer, each layer of the IC devices described herein may also include other types of devices besides logic or memory devices described herein. For example, in some embodiments, IC devices with logic devices incorporating transistors having back-side S/D contacts may also include memory cells in any of the layers.

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Furthermore, in the following detailed description, various aspects of the illustrative implementations will be described using terms commonly employed by those skilled in the art to convey the substance of their work to others skilled in the art.

For example, a term “interconnect” may be used to describe any element formed of an electrically conductive material for providing electrical connectivity to one or more components associated with an IC or/and between various such components. In general, the “interconnect” may refer to both conductive lines/wires (also sometimes referred to as “lines” or “metal lines” or “trenches”) and conductive vias (also sometimes referred to as “vias” or “metal vias”). In general, a term “conductive line” may be used to describe an electrically conductive element isolated by a dielectric material typically comprising an interlayer low-k dielectric that is provided within the plane of an IC chip. Such conductive lines are typically arranged in several levels, or several layers, of metallization stacks. On the other hand, the term “conductive via” may be used to describe an electrically conductive element that interconnects two or more conductive lines of different levels of a metallization stack. To that end, a via may be provided substantially perpendicularly to the plane of an IC chip or a support structure over which an IC structure is provided and may interconnect two conductive lines in adjacent levels or two conductive lines in not adjacent levels. A term “metallization stack” may be used to refer to a stack of one or more interconnects for providing connectivity to different circuit components of an IC chip.

In another example, the terms “package” and “IC package” are synonymous, as are the terms “die” and “IC die,” the term “insulating” means “electrically insulating,” the term “conducting” means “electrically conducting,” unless otherwise specified. Although certain elements may be referred to in the singular herein, such elements may include multiple sub-elements. For example, “an electrically conductive material” may include one or more electrically conductive materials. If used, the terms “oxide,” “carbide,” “nitride,” etc. refer to compounds containing, respectively, oxygen, carbon, nitrogen, etc., the term “high-k dielectric” refers to a material having a higher dielectric constant than silicon oxide, while the term “low-k dielectric” refers to a material having a lower dielectric constant than silicon oxide. Furthermore, the term “connected” may be used to describe a direct electrical or magnetic connection between the things that are connected, without any intermediary devices, while the term “coupled” may be used to describe either a direct electrical or magnetic connection between the things that are connected, or an indirect connection through one or more passive or active intermediary devices. The term “circuit” may be used to describe one or more passive and/or active components that are arranged to cooperate with one another to provide a desired function. The terms “substantially,” “close,” “approximately,” “near,” and “about,” generally refer to being within $\pm 20\%$ of a target value based on the context of a particular value as described herein or as known in the art. Similarly, terms indicating orientation of various elements, e.g., “coplanar,” “perpendicular,” “orthogonal,” “parallel,” or any other angle between the elements, generally refer to being within $\pm 5\text{--}20\%$ of a target value based on the context of a particular value as described herein or as known in the art.

For the purposes of the present disclosure, the phrase “A and/or B” means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The term “between,” when used with reference to

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measurement ranges, is inclusive of the ends of the measurement ranges. As used herein, the notation “A/B/C” means (A), (B), and/or (C).

The description may use the phrases “in an embodiment” or “in embodiments,” which may each refer to one or more of the same or different embodiments. Furthermore, the terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. The disclosure may use perspective-based descriptions such as “above,” “below,” “top,” “bottom,” and “side”; such descriptions are used to facilitate the discussion and are not intended to restrict the application of disclosed embodiments. The accompanying drawings are not necessarily drawn to scale. Unless otherwise specified, the use of the ordinal adjectives “first,” “second,” and “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking or in any other manner.

In the following detailed description, reference is made to the accompanying drawings that form a part hereof, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense. For convenience, if a collection of drawings designated with different letters are present, e.g., FIGS. 9A-9D, such a collection may be referred to herein without the letters, e.g., as “FIG. 9.”

In the drawings, some schematic illustrations of example structures of various devices and assemblies described herein may be shown with precise right angles and straight lines, but it is to be understood that such schematic illustrations may not reflect real-life process limitations which may cause the features to not look so “ideal” when any of the structures described herein are examined using e.g., scanning electron microscopy (SEM) images or transmission electron microscope (TEM) images. In such images of real structures, possible processing defects could also be visible, e.g., not-perfectly straight edges of materials, tapered vias or other openings, inadvertent rounding of corners or variations in thicknesses of different material layers, occasional screw, edge, or combination dislocations within the crystalline region, and/or occasional dislocation defects of single atoms or clusters of atoms. There may be other defects not listed here but that are common within the field of device fabrication.

Various operations may be described as multiple discrete actions or operations in turn in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed, and/or described operations may be omitted in additional embodiments.

Various IC assemblies with back-side power delivery and glass support at the front as described herein may be implemented in, or associated with, one or more components associated with an IC or/and may be implemented between various such components. In various embodiments, components associated with an IC include, for example, transistors, diodes, power sources, resistors, capacitors, inductors, sensors, transceivers, receivers, antennas, etc. Components

associated with an IC may include those that are mounted on IC or those connected to an IC. The IC may be either analog or digital and may be used in a number of applications, such as microprocessors, optoelectronics, logic blocks, audio amplifiers, etc., depending on the components associated with the IC. The IC may be employed as part of a chipset for executing one or more related functions in a computer.

Example Transistor Architectures

FIG. 1 provides a schematic illustration of a cross-sectional view of an example transistor **100**, implemented as a FET, with a back-side contact according to some embodiments of the present disclosure.

A number of elements labeled in FIG. 1 and in at least some of the subsequent figures with reference numerals are illustrated in these figures with different patterns, with a legend showing the correspondence between the reference numerals and patterns being provided at the bottom of each drawing page containing these figures. For example, the legend illustrates that FIG. 1 uses different patterns to show a channel material **102**, S/D regions **104**, contacts **104** to S/D regions, etc. Furthermore, although a certain number of a given element may be illustrated in FIG. 1 and in at least some of the subsequent figures, this is also simply for ease of illustration, and more, or less, than that number may be included in an IC device according to various embodiments of the present disclosure. Still further, various IC device views shown in FIG. 1 and in at least some of the subsequent figures are intended to show relative arrangements of various elements therein, and that various IC devices, or portions thereof, may include other elements or components that are not illustrated (e.g., any further materials, such as spacer materials that may surround the gate stack of the transistor **100**, etch-stop materials, etc.).

In general, a FET, e.g., a MOSFET, is a three-terminal device that includes source, drain, and gate terminals and uses electric field to control current flowing through the device. A FET typically includes a channel material, a source region and a drain regions provided in the channel material, and a gate stack that includes a gate electrode material, alternatively referred to as a “work function” (WF) material, provided over a portion of the channel material between the source and the drain regions, and, optionally, also includes a gate dielectric material between the gate electrode material and the channel material. This general structure is shown in FIG. 1, illustrating a channel material **102**, S/D regions **104** (shown as a first S/D region **104-1**, e.g., a source region, and a second S/D region **104-2**, e.g., a drain region), contacts **106** to S/D regions (shown as a first S/D contact **106-1**, providing electrical contact to the first S/D region **104-1**, and a second S/D contact **106-2**, providing electrical contact to the second S/D region **104-2**), and a gate stack **108**, which includes at least a gate electrode **110** and may also, optionally, include a gate dielectric **112**.

Implementations of the present disclosure may be formed or carried out on a support structure, which may be, e.g., a substrate, a die, a wafer or a chip. The substrate may, e.g., be the wafer **2000** of FIG. 11A, discussed below, and may be, or be included in, a die, e.g., the singulated die **2002** of FIG. 11B, discussed below. The substrate may be a semiconductor substrate composed of semiconductor material systems including, for example, N-type or P-type materials systems. In one implementation, the semiconductor substrate may be a crystalline substrate formed using a bulk silicon or a silicon-on-insulator (SOI) substructure. In other implementations, the semiconductor substrate may be formed using alternate materials, which may or may not be combined with silicon, that include, but are not limited to,

germanium, silicon germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, aluminum gallium arsenide, aluminum arsenide, indium aluminum arsenide, aluminum indium antimonide, indium gallium arsenide, gallium nitride, indium gallium nitride, aluminum indium nitride or gallium antimonide, or other combinations of group III-V materials (i.e., materials from groups III and V of the periodic system of elements), group II-VI (i.e., materials from groups II and IV of the periodic system of elements), or group IV materials (i.e., materials from group IV of the periodic system of elements). In some embodiments, the substrate may be non-crystalline. In some embodiments, the substrate may be a printed circuit board (PCB) substrate. Although a few examples of materials from which the substrate may be formed are described here, any material that may serve as a foundation upon which IC assemblies with back-side power delivery and glass support at the front as described herein may be built falls within the spirit and scope of the present disclosure. In various embodiments, the channel material **102** may include, or may be formed upon, any such substrate material that provides a suitable surface for forming the transistor **100**.

In some embodiments, the channel material **102** may be composed of semiconductor material systems including, for example, N-type or P-type materials systems. In some embodiments, the channel material **102** may include a high mobility oxide semiconductor material, such as tin oxide, antimony oxide, indium oxide, indium tin oxide, titanium oxide, zinc oxide, indium zinc oxide, gallium oxide, titanium oxynitride, ruthenium oxide, or tungsten oxide. In some embodiments, the channel material **102** may include a combination of semiconductor materials where one semiconductor material may be used for the channel portion (e.g., a portion **114** shown in FIG. 1, which is supposed to refer to the upper-most portion of the channel material **102**) and another material, sometimes referred to as a “blocking material,” may be used between the channel portion **114** and the support structure over which the transistor **100** is provided. In some embodiments, the channel material **102** may include a monocrystalline semiconductor, such as silicon (Si) or germanium (Ge). In some embodiments, the channel material **102** may include a compound semiconductor with a first sub-lattice of at least one element from group III of the periodic table (e.g., Al, Ga, In), and a second sub-lattice of at least one element of group V of the periodic table (e.g., P, As, Sb).

For some example N-type transistor embodiments (i.e., for the embodiments where the transistor **100** is an N-type metal oxide semiconductor (NMOS)), the channel portion **114** of the channel material **102** may advantageously include a III-V material having a high electron mobility, such as, but not limited to InGaAs, InP, InSb, and InAs. For some such embodiments, the channel portion **114** of the channel material **102** may be a ternary III-V alloy, such as InGaAs, GaAsSb, InAsP, or InPSb. For some $\text{In}_x\text{Ga}_{1-x}\text{As}$ fin embodiments, In content (x) may be between 0.6 and 0.9, and may advantageously be at least 0.7 (e.g., $\text{In}_{0.7}\text{Ga}_{0.3}\text{As}$). In some embodiments with highest mobility, the channel portion **114** of the channel material **102** may be an intrinsic III-V material, i.e., a III-V semiconductor material not intentionally doped with any electrically active impurity. In alternate embodiments, a nominal impurity dopant level may be present within the channel portion **114** of the channel material **102**, for example to further fine-tune a threshold voltage V_t , or to provide HALO pocket implants, etc. Even for impurity-doped embodiments however, impurity dopant level within the channel portion **114** of the channel material

102 may be relatively low, for example below 10^{15} dopant atoms per cubic centimeter (cm^{-3}), and advantageously below 10^{13} cm^{-3} .

For some example P-type transistor embodiments (i.e., for the embodiments where the transistor **100** is a P-type metal oxide semiconductor (PMOS)), the channel portion **114** of the channel material **102** may advantageously be a group IV material having a high hole mobility, such as, but not limited to Ge or a Ge-rich SiGe alloy. For some example embodiments, the channel portion **114** of the channel material **102** may have a Ge content between 0.6 and 0.9, and advantageously may be at least 0.7. In some embodiments with highest mobility, the channel portion **114** may be intrinsic III-V (or IV for P-type devices) material and not intentionally doped with any electrically active impurity. In alternate embodiments, one or more a nominal impurity dopant level may be present within the channel portion **114**, for example to further set a threshold voltage (V_t), or to provide HALO pocket implants, etc. Even for impurity-doped embodiments however, impurity dopant level within the channel portion is relatively low, for example below 10^{15} cm^{-3} , and advantageously below 10^{13} cm^{-3} .

In some embodiments, the transistor **100** may be a thin-film transistor (TFT). A TFT is a special kind of a field-effect transistor made by depositing a thin film of an active semiconductor material, as well as a dielectric layer and metallic contacts, over a supporting layer that may be a non-conducting layer. At least a portion of the active semiconductor material forms a channel of the TFT. If the transistor **100** is a TFT, the channel material **102** may include a high mobility oxide semiconductor material, such as tin oxide, antimony oxide, indium oxide, indium tin oxide, titanium oxide, zinc oxide, indium zinc oxide, indium gallium zinc oxide (IGZO), gallium oxide, titanium oxynitride, ruthenium oxide, or tungsten oxide. In general, if the transistor **100** is a TFT, the channel material **102** may include one or more of tin oxide, cobalt oxide, copper oxide, antimony oxide, ruthenium oxide, tungsten oxide, zinc oxide, gallium oxide, titanium oxide, indium oxide, titanium oxynitride, indium tin oxide, indium zinc oxide, nickel oxide, niobium oxide, copper peroxide, IGZO, indium telluride, molybdenite, molybdenum diselenide, tungsten diselenide, tungsten disulfide, N- or P-type amorphous or polycrystalline silicon, germanium, indium gallium arsenide, silicon germanium, gallium nitride, aluminum gallium nitride, indium phosphite, and black phosphorus, each of which may possibly be doped with one or more of gallium, indium, aluminum, fluorine, boron, phosphorus, arsenic, nitrogen, tantalum, tungsten, and magnesium, etc. In some embodiments, the channel material **102** may have a thickness between about 5 and 75 nanometers, including all values and ranges therein. In some embodiments, a thin film channel material **102** may be deposited at relatively low temperatures, which allows depositing the channel material **102** within the thermal budgets imposed on back end fabrication to avoid damaging other components, e.g., front end components such as the logic devices.

As shown in FIG. 1, a first and a second S/D regions **104-1**, **104-2** (together referred to as “S/D regions **104**”) may be included on either side of the gate stack **108**, thus realizing a transistor. As is known in the art, source and drain regions (also sometimes interchangeably referred to as “diffusion regions”) are formed for the gate stack of a FET. In some embodiments, the S/D regions **104** of the transistor **100** may be regions of doped semiconductors, e.g., regions of the channel material **102** (e.g., of the channel portion **114**) doped with a suitable dopant to a suitable dopant concen-

tration, so as to supply charge carriers for the transistor channel. In some embodiments, the S/D regions **104** may be highly doped, e.g., with dopant concentrations of about $1 \cdot 10^{21} \text{ cm}^{-3}$, in order to advantageously form Ohmic contacts with the respective S/D contacts **106**, although, in other embodiments, these regions may also have lower dopant concentrations and may form Schottky contacts in some implementations. Irrespective of the exact doping levels, the S/D regions **104** of the transistor **100** may be the regions having dopant concentration higher than in other regions, e.g., higher than a dopant concentration in a region of the channel material **102** between the first S/D region **104-1** and the second S/D region **104-2**, and, therefore, may be referred to as “highly doped” (HD) regions. In some embodiments, the S/D regions **104** may generally be formed using either an implantation/diffusion process or an etching/deposition process. In the former process, dopants such as boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the one or more semiconductor materials of the upper portion of the channel material **102** to form the S/D regions **104**. An annealing process that activates the dopants and causes them to diffuse further into the channel material **102** may follow the ion implantation process. In the latter process, the one or more semiconductor materials of the channel material **102** may first be etched to form recesses at the locations for the future S/D regions. An epitaxial deposition process may then be carried out to fill the recesses with material (which may include a combination of different materials) that is used to fabricate the S/D regions **104**. In some implementations, the S/D regions **104** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some implementations, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In further embodiments, the S/D regions **104** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. Although FIG. 1 illustrates the first and second S/D regions **104** with a single pattern, suggesting that the material composition of the first and second S/D regions **104** is the same, this may not be the case in some other embodiments of the transistor **100**. Thus, in some embodiments, the material composition of the first S/D region **104-1** may be different from the material composition of the second S/D region **104-2**.

As further shown in FIG. 1, S/D contacts **106-1** and **106-2** (together referred to as “S/D contacts **106**”), formed of one or more electrically conductive materials, may be used for providing electrical connectivity to the S/D regions **104-1** and **104-2**, respectively. In various embodiments, one or more layers of metal and/or metal alloys may be used to form the S/D contacts **106**. For example, the electrically conductive materials of the S/D contacts **106** may include one or more metals or metal alloys, with materials such as copper, ruthenium, palladium, platinum, cobalt, nickel, hafnium, zirconium, titanium, tantalum, and aluminum, tantalum nitride, tungsten, doped silicon, doped germanium, or alloys and mixtures of any of these. In some embodiments, the S/D contacts **106** may include one or more electrically conductive alloys, oxides, or carbides of one or more metals. In some embodiments, the S/D contacts **106** may include a doped semiconductor, such as silicon or another semiconductor doped with an N-type dopant or a P-type dopant. Metals may provide higher conductivity, while doped semiconductors may be easier to pattern during fabrication. Although FIG. 1 illustrates the first and second S/D contacts **106** with a single pattern, suggesting that the material composition of the first and second S/D contacts **106** is the

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same, this may not be the case in some other embodiments of the transistor **100**. Thus, in some embodiments, the material composition of the first S/D contact **106-1** may be different from the material composition of the second S/D contact **106-2**.

Turning to the gate stack **108**, the gate electrode **110** may include at least one P-type WF metal or N-type WF metal, depending on whether the transistor **100** is a PMOS transistor or an NMOS transistor. For a PMOS transistor, metals that may be used for the gate electrode **110** may include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, and conductive metal oxides (e.g., ruthenium oxide). For an NMOS transistor, metals that may be used for the gate electrode **110** include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, and carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide). In some embodiments, the gate electrode **110** may include a stack of two or more metal layers, where one or more metal layers are WF metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included for other purposes, such as to act as a diffusion barrier layer, described below.

If used, the gate dielectric **112** may at least laterally surround the channel portion **114**, and the gate electrode **110** may laterally surround the gate dielectric **112** such that the gate dielectric **112** is disposed between the gate electrode **110** and the channel material **104**. In various embodiments, the gate dielectric **112** may include one or more high-k dielectric materials and may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Examples of high-k materials that may be used in the gate dielectric **112** may include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, tantalum oxide, tantalum silicon oxide, lead scandium tantalum oxide, and lead zinc niobate. In some embodiments, an annealing process may be carried out on the gate dielectric **112** during manufacture of the transistor **100** to improve the quality of the gate dielectric **112**. In some embodiments, the gate dielectric **112** may have a thickness between about 0.5 nanometers and 3 nanometers, including all values and ranges therein, e.g., between about 1 and 3 nanometers, or between about 1 and 2 nanometers.

In some embodiments, the gate dielectric **112** may be a multilayer gate dielectric, e.g., it may include any of the high-k dielectric materials in one layer and a layer of IGZO. In some embodiments, the gate stack **108** may be arranged so that the IGZO is disposed between the high-k dielectric and the channel material **104**. In such embodiments, the IGZO may be in contact with the channel material **104** and may provide the interface between the channel material **104** and the remainder of the multilayer gate dielectric **112**. The IGZO may have a gallium to indium ratio of 1:1, a gallium to indium ratio greater than 1 (e.g., 2:1, 3:1, 4:1, 5:1, 6:1, 7:1, 8:1, 9:1, or 10:1), and/or a gallium to indium ratio less than 1 (e.g., 1:2, 1:3, 1:4, 1:5, 1:6, 1:7, 1:8, 1:9, or 1:10).

In some embodiments, the gate stack **108** may be surrounded by a dielectric spacer, not specifically shown in FIG. 1. The dielectric spacer may be configured to provide separation between the gate stacks **108** of different transistors **100** which may be provided adjacent to one another (e.g., different transistors **100** provided along a single fin if

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the transistors **100** are FinFETs), as well as between the gate stack **108** and one of the S/D contacts **106** that is disposed on the same side as the gate stack **108**. Such a dielectric spacer may include one or more low-k dielectric materials.

Examples of the low-k dielectric materials that may be used as the dielectric spacer include, but are not limited to, silicon dioxide, carbon-doped oxide, silicon nitride, fused silica glass (FSG), and organosilicates such as silsesquioxane, siloxane, and organosilicate glass. Other examples of low-k dielectric materials that may be used as the dielectric spacer include organic polymers such as polyimide, polynorbornenes, benzocyclobutene, perfluorocyclobutane, or polytetrafluoroethylene (PTFE). Still other examples of low-k dielectric materials that may be used as the dielectric spacer include silicon-based polymeric dielectrics such as hydrogen silsesquioxane (HSQ) and methylsilsesquioxane (MSQ). Other examples of low-k materials that may be used in a dielectric spacer include various porous dielectric materials, such as for example porous silicon dioxide or porous carbon-doped silicon dioxide, where large voids or pores are created in a dielectric in order to reduce the overall dielectric constant of the layer, since voids can have a dielectric constant of nearly 1.

In stark contrast to conventional implementations where both S/D contacts are typically provided on a single side of a transistor, typically on the front side, e.g., where the gate stack **108** is provided, the two S/D contacts **106** are provided on different sides. Namely, as shown in FIG. 1, the second S/D contact **106-2** is provided on the same side as the gate stack **108**, which may be considered to be the front side of the transistor **100**, while the first S/D contact **106-1** is provided on the opposite side, which may be considered to be the back side of the transistor **100**. Thus, the first S/D contact **106-1** is the back-side contact and the second S/D contact **106-2** is the front-side contact of the transistor **100**. If considering the layers above a support structure (not shown in FIG. 1) over which the entire transistor **100** is built, then the first S/D contact **106-1** may be considered to be in a first layer **120-1** above the support structure, the second S/D contact **106-2** may be considered to be in a second layer **120-2** above the support structure, and a portion of the channel material **102** between the first S/D region **104-1** and the second S/D region **104-2** (e.g., the channel portion **114**) is in a third layer **120-3** over the support structure. As can be seen from FIG. 1, the third layer **120-3** is between the first layer **120-1** and the second layer **120-2**. At least a portion of the gate stack **108**, or a contact to the gate stack **108** (such as a gate contact not specifically shown in FIG. 1), may be provided in the same layer as one of the S/D contacts **106**, e.g., in the second layer **120-2**, as shown in FIG. 1. In further embodiments of the transistor **100**, the first S/D contact **106-1** may also be implemented in the second layer **120-2**.

Transistors with back-side S/D contacts as described herein, such as the transistor **100**, may be implemented using any suitable transistor architecture, e.g., planar or non-planar architectures. One example structure is shown in FIGS. 2A-2B, illustrating perspective and cross-sectional views, respectively, of an example IC device **200** having a transistor with at least one back-side contact implemented as a FinFET, according to some embodiments of the present disclosure. Thus, the IC device **200** illustrates one example implementation of the transistor **100**. Therefore, some of the reference numerals shown in FIGS. 2A-2B are the same as those used in FIG. 1, indicating the same or similar elements as those described with reference to FIG. 1, so that their descriptions are not repeated for FIGS. 2A-2B.

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FinFETs refer to transistors having a non-planar architecture where a fin, formed of one or more semiconductor materials, extends away from a base (where the term “base” refers to any suitable support structure on which a transistor may be built, e.g., a substrate). A portion of the fin that is closest to the base may be enclosed by an insulator material. Such an insulator material, typically an oxide, is commonly referred to as a “shallow trench isolation” (STI), and the portion of the fin enclosed by the STI is typically referred to as a “subfin portion” or simply a “subfin.” A gate stack that includes at least a layer of a gate electrode material and, optionally, a layer of a gate dielectric may be provided over the top and sides of the remaining upper portion of the fin (i.e., the portion above and not enclosed by the STI), thus wrapping around the upper-most portion of the fin. The portion of the fin over which the gate stack wraps around is typically referred to as a “channel portion” of the fin because this is where, during operation of the transistor, a conductive channel forms, and is a part of an active region of the fin. A source region and a drain region are provided on the opposite sides of the gate stack, forming, respectively, a source and a drain terminal of a transistor. FinFETs may be implemented as “tri-gate transistors,” where the name “tri-gate” originates from the fact that, in use, such transistors may form conducting channels on three “sides” of the fin. FinFETs potentially improve performance relative to single-gate transistors and double-gate transistors.

FIG. 2A is a perspective view, while FIG. 2B is a cross-sectional side view of an IC device/Fin FET **200** with one front-side and one back-side S/D contact, according to some embodiments of the disclosure. FIGS. 2A-2B illustrate the channel material **102**, the S/D regions **104**, and the gate stack **108** showing the gate electrode **110** and the gate dielectric **112** as described above. As shown in FIGS. 2A-2B, when the transistor **100** is implemented as a FinFET, the FinFET **200** may further include a base **202**, a fin **204**, and an STI material **206** enclosing the subfin portion of the fin **204**. The S/D contacts **106** are not specifically shown in FIGS. 2A-2B in order to not clutter the drawings. The cross-sectional side view of FIG. 2B is the view in the y-z plane of the example coordinate system x-y-z shown in FIG. 2A, with the cross-section of FIG. 2B taken across the fin **204** (e.g., along the plane shown in FIG. 2A as a plane AA). On the other hand, the cross-sectional side view of FIG. 1 is the view in the x-z plane of the example coordinate system shown in FIG. 2A with the cross-section taken along the fin **204** for one example portion of the gate stack **108** (e.g., along the plane shown in FIG. 2A and in FIG. 2B as a plane BB).

As shown in FIGS. 2A-2B, the fin **204** may extend away from the base **202** and may be substantially perpendicular to the base **202**. The fin **204** may include one or more semiconductor materials, e.g., a stack of semiconductor materials, so that the upper-most portion of the fin (namely, the portion of the fin **204** enclosed by the gate stack **108**) may serve as the channel region of the FinFET **200**. Therefore, the upper-most portion of the fin **204** may be formed of the channel material **102** as described above and may include the channel portion **114**.

The subfin of the fin **204** may be a binary, ternary, or quaternary III-V compound semiconductor that is an alloy of two, three, or even four elements from groups III and V of the periodic table, including boron, aluminum, indium, gallium, nitrogen, arsenic, phosphorus, antimony, and bismuth. For some example N-type transistor embodiments, the subfin portion of the fin **204** may be a III-V material having a band offset (e.g., conduction band offset for N-type

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devices) from the channel portion. Example materials, include, but are not limited to, GaAs, GaSb, GaAsSb, GaP, InAlAs, GaAsSb, AlAs, AlP, AlSb, and AlGaAs. In some N-type transistor embodiments of the FinFET **200** where the channel portion of the fin **204** (e.g., the channel portion **114**) is InGaAs, the subfin may be GaAs, and at least a portion of the subfin may also be doped with impurities (e.g., P-type) to a greater impurity level than the channel portion. In an alternate heterojunction embodiment, the subfin and the channel portion of the fin **204** are each, or include, group IV semiconductors (e.g., Si, Ge, SiGe). The subfin of the fin **204** may be a first elemental semiconductor (e.g., Si or Ge) or a first SiGe alloy (e.g., having a wide bandgap). For some example P-type transistor embodiments, the subfin of the fin **204** may be a group IV material having a band offset (e.g., valance band offset for P-type devices) from the channel portion. Example materials, include, but are not limited to, Si or Si-rich SiGe. In some P-type transistor embodiments, the subfin of the fin **204** is Si and at least a portion of the subfin may also be doped with impurities (e.g., N-type) to a higher impurity level than the channel portion.

As further shown in FIGS. 2A-2B, the STI material **206** may enclose portions of the sides of the fin **204**. A portion of the fin **204** enclosed by the STI **106** forms a subfin. In various embodiments, the STI material **206** may be a low-k or high-k dielectric including, but not limited to, elements such as hafnium, silicon, oxygen, nitrogen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Further examples of dielectric materials that may be used in the STI material **206** may include, but are not limited to, silicon nitride, silicon oxide, silicon dioxide, silicon carbide, silicon nitride doped with carbon, silicon oxynitride, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, tantalum oxide, tantalum silicon oxide, lead scandium tantalum oxide, and lead zinc niobate.

The gate stack **108** may wrap around the upper portion of the fin **204** (the portion above the STI **206**), as shown in FIGS. 2A-2B, with a channel portion of the fin **204** (e.g., the channel portion **114**, described above) corresponding to the portion of the fin **204** wrapped by the gate stack **108** as shown in FIGS. 2A-2B. In particular, the gate dielectric **112** (if used) may wrap around the upper-most portion of the fin **204**, and the gate electrode **110** may wrap around the gate dielectric **112**. The interface between the channel portion and the subfin portion of the fin **204** is located proximate to where the gate electrode **110** ends.

In some embodiments, the FinFET **200** may have a gate length, GL, (i.e., a distance between the first S/D region **104-1** and the second S/D region **104-2**), a dimension measured along the fin **204** in the direction of the x-axis of the example reference coordinate system x-y-z shown in FIG. 1 and FIGS. 2A-2B, which may, in some embodiments, be between about 5 and 40 nanometers, including all values and ranges therein (e.g., between about 22 and 35 nanometers, or between about 20 and 30 nanometers). The fin **204** may have a thickness, a dimension measured in the direction of the y-axis of the reference coordinate system x-y-z shown in FIGS. 2A-2B, that may, in some embodiments, be between about 5 and 30 nanometers, including all values and ranges therein (e.g., between about 7 and 20 nanometers, or between about 10 and 15 nanometers). The fin **204** may have a height, a dimension measured in the direction of the z-axis

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of the reference coordinate system x-y-z shown in FIG. 1, which may, in some embodiments, be between about 30 and 350 nanometers, including all values and ranges therein (e.g., between about 30 and 200 nanometers, between about 75 and 250 nanometers, or between about 150 and 300 nanometers).

Although the fin **204** illustrated in FIGS. 2A-2B is shown as having a rectangular cross-section in a y-z plane of the reference coordinate system shown, the fin **204** may instead have a cross-section that is rounded or sloped at the “top” of the fin **204**, and the gate stack **108** may conform to this rounded or sloped fin **204**. In use, the FinFET **200** may form conducting channels on three “sides” of the channel portion of the fin **204**, potentially improving performance relative to single-gate transistors (which may form conducting channels on one “side” of a channel material or substrate) and double-gate transistors (which may form conducting channels on two “sides” of a channel material or substrate).

While not specifically shown in FIG. 2A, S/D contacts **106** may be electrically connected to the S/D regions **104**, but extending in different vertical directions with respect to the fin **204**. For example, the first S/D contact **106-1** may be electrically connected to the first S/D region **104-1** and extend from the first S/D region **104-1** towards the base **202**, thus forming a back-side S/D contact for the FinFET **200**, similar to the illustration of FIG. 1. In such implementation, the second S/D contact **106-2** may be electrically connected to the second S/D region **104-2** and extend from the second S/D region **104-2** away from the base **202**, thus forming a front-side S/D contact for the FinFET **200**, also similar to the illustration of FIG. 1.

While FIGS. 2A-2B illustrate a single FinFET **200**, in some embodiments, a plurality of FinFETs may be arranged next to one another (with some spacing in between) along the fin **204**. Furthermore, in various further embodiments, the transistor **100** with one front-side and one back-side S/D contacts may be implemented in many other transistor architectures besides the FinFET **200**, such as planar FETs, nanowire FETs, or nanoribbon FETs.

Example Memory Implementations

Embedded memory is important to the performance of modern system-on-a-chip (SoC) technology, especially to enable 3D monolithic integration. IC assemblies with back-side power delivery and glass support at the front may include embedded memory. Therefore, some considerations related to memory will now be described.

Some memory devices may be considered “standalone” devices in that they are included in a chip that does not also include compute logic (where, as used herein, the term “compute logic devices” or simply “compute logic” or “logic devices,” refers to devices, e.g., transistors, for performing computing/processing operations). Other memory devices may be included in a chip along with compute logic and may be referred to as “embedded” memory devices. Using embedded memory to support compute logic may improve performance by bringing the memory and the compute logic closer together and eliminating interfaces that increase latency. Various embodiments of the present disclosure relate to embedded memory arrays, as well as corresponding methods and devices.

Some embodiments of the present disclosure may refer to dynamic random-access memory (DRAM) and in particular, embedded DRAM (eDRAM), because this type of memory has been introduced in the past to address the limitation in density and standby power of large static random-access memory (SRAM)-based caches. However, embodiments of the present disclosure are equally applicable to memory cells

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implemented other technologies. Thus, in general, memory cells described herein may be implemented as eDRAM cells, spin-transfer torque random-access memory (STTRAM) cells, resistive random-access memory (RRAM) cells, or any other nonvolatile memory cells.

A memory cell, e.g., an eDRAM cell, may include a capacitor for storing a bit value, or a memory state (e.g., logical “1” or “0”) of the cell, and an access transistor controlling access to the cell (e.g., access to write information to the cell or access to read information from the cell). Such a memory cell may be referred to as a “1T-1C memory cell,” highlighting the fact that it uses one transistor (i.e., “1T” in the term “1T-1C memory cell”) and one capacitor (i.e., “1C” in the term “1T-1C memory cell”). The capacitor of a 1T-1C memory cell may be coupled to one source/drain (S/D) terminal of the access transistor (e.g., to the source terminal of the access transistor), while the other S/D terminal of the access transistor may be coupled to a bitline (BL), and a gate terminal of the transistor may be coupled to a wordline (WL). Since such a memory cell can be fabricated with as little as a single access transistor, it can provide higher density and lower standby power versus SRAM in the same process technology.

Various 1T-1C memory cells have, conventionally, been implemented with access transistors being FEOL, logic-process based, transistors implemented in an upper-most layer of a semiconductor substrate. Inventors of the present disclosure realized that using conventional logic transistors creates several challenges if such transistors are to be used to create three-dimensional memory and logic devices.

One challenge relates to the location of the capacitors such memory cells. Namely, it may be desirable to provide capacitors in metal layers close to their corresponding access transistors. Since logic transistors are implemented as FEOL transistors provided directly on the semiconductor substrate, the corresponding capacitors of 1T-1C memory cells then have to be embedded in lower metal layers in order to be close enough to the logic access transistors. As the pitches of lower metal layers aggressively scale in advanced technology nodes, embedding the capacitors in the lower metal layers poses significant challenges to the scaling of 1T-1C based memory and to creation of three-dimensional memory devices.

Another challenge resides in that, given a usable surface area of a substrate, there are only so many FEOL transistors that can be formed in that area, placing a significant limitation on the density of memory cells or logic devices incorporating such transistors.

Implementing transistors of memory cells (e.g., access transistors of memory cells) as transistors with back-side contacts may improve on at least some of the challenges and issues described above. For example, moving access transistors of memory cells to the BEOL layers (enabled by the back-side contact architecture) means that their corresponding capacitors can be implemented in the upper metal layers with correspondingly thicker interlayer dielectric (ILD) and larger metal pitch to achieve higher capacitance, which may ease the integration challenge introduced by embedding the capacitors.

FIG. 3 provides a schematic illustration of a cross-sectional view of an example memory cell **300** that includes a transistor with a back-side contact, according to some embodiments of the present disclosure. FIG. 3 illustrates how the transistor **100** may be used to form a 1T-1C memory cell. In particular, the memory cell **300** illustrates all of the components of the transistor **100** of FIG. 1 (the descriptions of which, therefore, not repeated here), and further sche-

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matically illustrates that, in some embodiments, a capacitor **302** may be coupled to the back-side S/D contact **106-1** of the transistor **100**. The capacitor **302** may be any suitable capacitor, e.g., a metal-insulator-metal (MIM) capacitor for storing a bit value, or a memory state (e.g., logical “1” or “0”) of the memory cell **300**, and the transistor **100** may then function as an access transistor controlling access to the memory cell **300** (e.g., access to write information to the cell or access to read information from the cell. By coupling the capacitor **302** to the S/D region **104-1**, the capacitor **302** is configured to store the memory state of the memory cell **300**. In some embodiments, the capacitor **302** may be coupled to the S/D region **104-1** via a storage node (not specifically shown in FIG. 3) coupled to the S/D region **104-1**. In some embodiments, the S/D contact **106-1** may be considered to be the storage node.

Although not specifically shown in FIG. 3, the memory cell **300** may further include a bitline to transfer the memory state and coupled to the one of the S/D regions **104** to which the capacitor **302** is not coupled (e.g., to the S/D region **104-2**, for the illustration of FIG. 3). Such a bitline can be connected to a sense amplifier and a bitline driver which may, e.g., be provided in a memory peripheral circuit associated with a memory array in which the memory cell **300** may be included. Furthermore, although also not specifically shown in FIG. 3, the memory cell **300** may further include a wordline, coupled to the gate terminal of the transistor **100**, e.g., coupled to the gate stack **108**, to supply a gate signal. The transistor **100** may be configured to control transfer of a memory state of the memory cell **300** between the bitline and the storage node or the capacitor **302** in response to the gate signal.

Example IC Assemblies with Glass Support at the Front

Transistors with back-side contacts may enable 3D integration of IC assemblies with back-side power delivery and glass support at the front. An example IC assembly is shown in FIG. 4, providing a block diagram of an IC assembly **400** with back-side power delivery and glass support at the front, according to some embodiments of the present disclosure.

As shown in FIG. 4, the IC assembly **400** may include an FEOL layer **420** and a BEOL layer **430** above FEOL layer **420**. The FEOL layer **420** may include a plurality of FEOL devices, e.g., FEOL transistors implemented as transistors with back-side contacts. The BEOL layer **430** may include at least a plurality of interconnects electrically coupled to (e.g., in electrically conductive contact with at least portions of) one or more of the plurality of FEOL devices of the FEOL layer **420**. In some embodiments, the BEOL layer **430** may further include BEOL devices, e.g., backend transistors, at least some of which may be implemented as transistors with back-side contacts.

In various embodiments, the FEOL transistors with back-side contacts, implemented in the FEOL layer **420**, may be a part of compute logic and/or a part of a memory array.

For example, in some embodiments, some of the FEOL transistors of the FEOL layer **420** may be access transistors of memory cells of a memory array, e.g., of the 1T-1C memory cells as described above. In such embodiments, the capacitors of such memory cells may then be implemented in the BEOL layer **430**. In other embodiments, some of the FEOL transistors of the FEOL layer **420** may be access transistors of memory cells of a type other than 1T-1C. In such embodiments, other portions of the memory cells (e.g., storage transistors) may be implemented in the BEOL layer **430**.

In another example, some of the FEOL transistors of the FEOL layer **420** may be a part of compute logic of the IC

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assembly **400**. For example, such transistors may be responsible for compute logic functionality related to read/write operations with respect to the data stored in the memory cells that may be implemented in the BEOL layer **430**. To that end, some of the FEOL transistors of the FEOL layer **420** may be part of one or more input/output (I/O) ICs (e.g., a memory peripheral circuit) configured to control (e.g., control access (read/write), store, refresh) the memory cells implemented in the IC assembly **400** (e.g., memory cells implemented in the BEOL layer **430**). In some embodiments, some of the FEOL transistors of the FEOL layer **420** may be part of high-performance compute logic, configured to perform various operations with respect to data stored in the memory cells implemented in the IC assembly **400** (e.g., arithmetic and logic operations, pipelining of data from one or more of the memory arrays implemented in the IC assembly **400**, and possibly also data from external devices/chips).

Transistors with back-side contacts as described herein, either as a standalone transistors (e.g., the transistor **100**) or included as a part of a memory cell (e.g., the memory cell **300**), may be included in various regions/locations in the IC assembly **400**. For example, the transistor **100** may be used as, e.g., a logic transistor in the compute logic (e.g., included in the FEOL layer **420**). In another example, the transistor **100** may be used as, e.g., access transistors in the one or more memory layers of the BEOL **430**. Providing transistors with back-side contacts may ease the integration challenge introduced by embedding the storage nodes (e.g., storage capacitors) of memory cells, and make building of three-dimensional memory and logic devices with a stacked architecture with many layers of memory and/or compute logic feasible.

The illustration of FIG. 4 is intended to provide a general orientation and arrangement of various layers with respect to one another, and, unless specified otherwise in the present disclosure, includes embodiments of the IC device **400** where portions of elements described with respect to one of the layers shown in FIG. 4 may extend into one or more, or be present in, other layers. For example, power and signal interconnects for the various IC components of the IC assembly **400** may be present in any of the layers shown in FIG. 4, although not specifically illustrated in FIG. 4. Furthermore, although a single BEOL layer **430** is shown in FIG. 4, in various embodiments, the BEOL layer **430** of the IC assembly **400** may include a plurality of BEOL layers.

In some embodiments, the BEOL layer **430** may include one or more memory layers which may form one or more memory arrays. Such a memory array may include access transistors (e.g., the transistor **100**), storage nodes (e.g., storage capacitors or storage transistors), as well as wordlines (e.g., row selectors) and bitlines (e.g., column selectors), making up memory cells. In some embodiments, the memory layers of the BEOL layer **430** may include TFT-based memory cells. On the other hand, the FEOL layer **420** may include various logic layers, circuits, and devices (e.g., logic transistors) to drive and control a logic IC. For example, the logic devices of the FEOL layer **420** may form a memory peripheral circuit to control (e.g., access (read/write), store, refresh) the memory cells of the BEOL layer **430**. In some embodiments of the IC assembly **400**, the compute logic may be provided in the FEOL **420** and in one or more lowest metal layer of the BEOL layer **430**, while the one or more memory arrays may be provided in higher layers of the BEOL layer **430**. In other embodiments of the IC assembly **400**, the compute logic described with reference to the FEOL layer **420** may be provided above the

FEOL layer **420** (e.g., in the BEOL layer **430**), in between memory layers of the BEOL layer **430**, or combined with the memory layers of the BEOL layer **430**.

Various BEOL layers of the BEOL layer **430** may be/include metal layers of a metallization stack of the IC assembly **400**. Various metal layers of the BEOL may be used to interconnect the various inputs and outputs of the logic devices in the compute logic of the FEOL layer **420** and/or of the memory cells in the memory layers of the BEOL layer **430**. Generally speaking, each of the metal layers of the BEOL layer **430** may include a via portion and a trench/interconnect portion. The trench portion of a metal layer is configured for transferring signals and power along electrically conductive (e.g., metal) lines (also sometimes referred to as “trenches”) extending in the x-y plane (e.g., in the x or y directions), while the via portion of a metal layer is configured for transferring signals and power through electrically conductive vias extending in the z-direction, e.g., to any of the adjacent metal layers above or below. Accordingly, vias connect metal structures (e.g., metal lines or vias) from one metal layer to metal structures of an adjacent metal layer. While referred to as “metal” layers, various layers of the BEOL layer **430** may include only certain patterns of conductive metals, e.g., copper (Cu), aluminum (Al), tungsten (W), or cobalt (Co), or metal alloys, or more generally, patterns of an electrically conductive material, formed in an insulating medium such as an ILD. The insulating medium may include any suitable ILD materials such as silicon oxide, carbon-doped silicon oxide, silicon carbide, silicon nitride, aluminum oxide, and/or silicon oxynitride.

The FEOL layer **420** may originally be provided over a semiconductor support structure such as a substrate, a die, a wafer or a chip, and may include any of the materials, or combinations of materials, described with reference to the support structure for the embodiments of FIGS. 1-3. However, such a semiconductor support structure may subsequently be removed to expose the back-side portions of the FEOL devices of the FEOL layer **420** so that a back-side power delivery structure **410** may be provided at the back side of the FEOL layer **420** (thus, the BEOL layer **430** is provided at the front side of the FEOL layer **420** and the back-side power delivery structure **410** is provided at the back side of the FEOL layer **420**).

As also shown in FIG. 4, the IC assembly **400** may further include a bonding interface **440** and a glass support structure **450**, where the bonding interface **440** may be an interface where a top face of the BEOL layer **430** is bonded to a face of the glass support structure **450**. Thus, in the IC assembly **400**, the FEOL layer **420** is between the back-side power delivery structure **410** and the BEOL layer **430**, and the BEOL layer **430** is between the FEOL layer **420** and the glass support structure **450**.

FIGS. 5-8 provide schematic illustrations of example implementations of the IC assembly **400**, according to various embodiments of the present disclosure.

FIG. 5 provides a schematic illustration of an IC assembly **500** with back-side power delivery and glass support at the front, according to some embodiments of the present disclosure. Portions of the IC assembly **400** that were shown in FIG. 4, such as the back-side power delivery structure **410**, the FEOL **420**, etc., are labeled in the IC assembly **500** of FIG. 5. The IC assembly **500** further illustrates example implementations of each of these portions.

As shown in FIG. 5, the back-side power delivery structure **410** may include a plurality of power interconnects **512**, arranged in one or more layers (three such layers are shown in FIG. 5, separated by horizontal lines, but such separation

may be absent in other embodiments, or other embodiments may include different number of layers and/or different arrangements of the power interconnects **512** than what is shown in FIG. 5). The power interconnects **512** may include any suitable combination of vias **512-1** and lines **512-2**, some of which are labeled in FIG. 5 and some of which are not labeled in order to not clutter the drawing. The power interconnects **512** may include any suitable conductive materials, such as any of the conductive metals or metal alloys as described above. Portions of various ones of the power interconnects **512** may be enclosed by an insulator material **514**, which may include any of the ILD materials described above.

As further shown in FIG. 5, the FEOL layer **420** may include a plurality of FEOL devices **526**. One or more of the FEOL devices **526** may be transistors with back-side contacts as described above, e.g., the transistors **100**. In various embodiments, the FEOL devices **526** may include one or more of fin-based transistors, nanoribbon transistors, and nanowire transistors as known in the art, but with one or more back-side contacts as described herein. One or more of the power interconnects **512** may then be coupled to one or more S/D regions of such transistors with back-side contacts (i.e., one or more power interconnects **512** may form back-side contacts to one or more S/D regions of the transistors of the FEOL devices **526**).

As also shown in FIG. 5, the BEOL layer **430** may include a plurality of BEOL interconnects **532**, which may include any suitable conductive materials, such as any of the conductive metals or metal alloys as described above. The BEOL interconnects **532** may include any suitable combination of vias **532-1** and lines **532-2**, some of which are labeled in FIG. 5 and some of which are not labeled in order to not clutter the drawing. One or more of the BEOL interconnects **532** may be electrically coupled to (e.g., in electrically conductive contact with at least portions of) one or more of the plurality of FEOL devices **526**. At least portions of the BEOL interconnects **532** may be enclosed by an insulator material **534**, which may include any of the ILD materials described above. In some embodiments, an insulator material such as the insulator material **534** may also at least partially enclose portions of the FEOL devices **526**. FIG. 5 also schematically illustrates that the BEOL layer **430** may include a layer of memory cells **536**. The memory cells **536** may be any of the memory cells described above, e.g., TFT-based memory cells, e.g., the memory cells **300**. In further embodiments, the IC assembly **500** may include multiple layers of the memory cells **536**.

In some embodiments, the cross-sectional side views of the BEOL interconnects **532** and the power interconnects **512** may have characteristic differences due to the fact that the BEOL interconnects **532** and the power interconnects **512** are formed on different sides of the FEOL layer **420**. In particular, in such embodiments, cross-sections, in a plane perpendicular to the FEOL layer **420**, of at least some of the BEOL interconnects **532** and at least some of the power interconnects **512** may be trapezoids. Such trapezoids may include two parallel sides, one of which is a short side and one of which is a long side (i.e., the length of the long side is greater than the length of the short side). Characteristic to the fact that the BEOL interconnects **532** and the power interconnects **512** are formed on different sides of the FEOL layer **420**, for the trapezoids of the BEOL interconnects **532**, the long side would be closer to the glass support structure **450** than the short side and the short side would be closer to the FEOL layer **420** than the long side, whereas, for the trapezoids of the power interconnects **512**, the short side

would be closer to both the glass support structure **450** and the FEOL layer **420** than the long side.

FIG. **5** further illustrates a bonding interface material **540**, which may be used to implement the bonding interface **440**, bonding the upper face of the BEOL layer **430** to a non-semiconductor support structure **550** that may be used to implement the glass support structure **450** as described above. In some embodiments, the bonding interface material **540** may include an oxide, e.g., silicon oxide. As shown in FIG. **5**, in some embodiments, portions (e.g., one face) of the bonding interface material **540** may be in contact with one or more portions of the glass support structure **450**, while other portions (e.g., the opposing face) of the bonding interface material **540** may be in contact with one or more portions of the BEOL layer **430**. In some embodiments, the bonding interface material **540** may have a thickness between about 1 nanometer and 100 nanometers, e.g., between about 1 and 50 nanometers, or between about 1 and 20 nanometers.

In some embodiments, the non-semiconductor support structure **550** may include a glass material. Examples of glass materials include silicon oxide materials, possibly doped with elements and compounds such as boron, carbon, aluminum, hafnium oxide, e.g., in doping concentrations of between about 0.01% and 10%. In other embodiments, the non-semiconductor support structure **550** may include other solid materials having a dielectric constant lower than that of Si, e.g., lower than about 10.5. In some embodiments, the non-semiconductor support structure **550** may include mica. A thickness of the glass support structure **450** may be of any value for the glass support structure **450** to provide mechanical stability for the IC assembly **400** and, possibly, to support inclusion of various devices for further reducing the parasitic effects in the IC assembly (some such devices being shown in FIG. **6** and described above). In some embodiments, the glass support structure **450** may have a thickness between about 0.2 micrometer (micron) and 100 micron, e.g., between about 0.5 and 5 micron, or between about 1 and 3 micron.

FIG. **6** provides a schematic illustration of an IC assembly **600** with back-side power delivery and glass support with thin-film devices at the front, according to some embodiments of the present disclosure. Portions of the IC assembly **400** that were shown in FIG. **4**, such as the back-side power delivery structure **410**, the FEOL **420**, etc., are labeled in the IC assembly **600** of FIG. **6**. The IC assembly **600** further illustrates example implementations of each of these portions. In particular, the IC assembly **600** may be implemented as the IC assembly **500** as described above (which is shown in FIG. **6** with the IC assembly **600** including some of the same elements, showing using the same patterns, as those included in the IC assembly **500** illustrated in FIG. **5**), except that the IC assembly **600** further includes one or more thin-film devices **556** disposed in the glass support structure **450**. In the interests of brevity, detailed descriptions of the IC assembly **500** are not repeated with respect to the IC assembly **600**, and only the differences are described. Furthermore, in order to not clutter the drawing of FIG. **6**, the power interconnects **512-1** and **512-2** and the BEOL interconnects **532-1** and **532-2** are not specifically labeled in FIG. **6** as they were labeled in FIG. **5**.

In various embodiments, the thin-film devices **556** may be two-terminal devices such as thin-film resistors, thin-film capacitors, and thin-film inductors, configured to reduce the parasitic effects within the IC assembly **600**. The first terminal of such a two-terminal thin-film device **556** may be electrically coupled (e.g., in electrically conductive contact

with) to a first BEOL interconnect of the plurality of BEOL interconnects **532**, while the second terminal may be electrically coupled (e.g., in electrically conductive contact with) to a second BEOL interconnect of the plurality of BEOL interconnects **532**. One example of such two-terminal coupling is labeled in FIG. **6** for one of the thin-film devices **556** (although three different thin-film devices **556** are shown in the example of FIG. **6**) with the first terminal of the thin-film device **556** shown on the right side of the IC assembly **600** being coupled to a first BEOL interconnect **612-1** of the plurality of BEOL interconnects **532** (said coupling shown in FIG. **6** within a dotted contour **652-1**), and with the second terminal of the thin-film device **556** shown on the right side of the IC assembly **600** being coupled to a second BEOL interconnect **612-2** of the plurality of BEOL interconnects **532** (said coupling shown in FIG. **6** within a dotted contour **652-2**). As shown in FIG. **6**, in some embodiments, portions of the thin-film devices **556** may extend through the bonding interface **450** to make electrical contact to respective portions of the BEOL interconnects **532**.

FIG. **7** provides a schematic illustration of an IC assembly **700** with back-side power delivery and glass support with an active layer at the front, according to some embodiments of the present disclosure. Portions of the IC assembly **400** that were shown in FIG. **4**, such as the back-side power delivery structure **410**, the FEOL **420**, etc., are labeled in the IC assembly **700** of FIG. **7**. The IC assembly **700** further illustrates example implementations of each of these portions. In particular, the IC assembly **700** may be implemented as the IC assembly **500** as described above (which is shown in FIG. **7** with the IC assembly **700** including some of the same elements, showing using the same patterns, as those included in the IC assembly **500** illustrated in FIG. **5**), except that the IC assembly **700** further includes an active layer **650** between the glass support structure **450** and the BEOL layer **430**. In the interests of brevity, detailed descriptions of the IC assembly **500** are not repeated with respect to the IC assembly **700**, and only the differences are described. Furthermore, in order to not clutter the drawing of FIG. **7**, the power interconnects **512-1** and **512-2** and the BEOL interconnects **532-1** and **532-2** are not specifically labeled in FIG. **7** as they were labeled in FIG. **5**.

As shown in FIG. **7**, the active layer **750** may be provided between the glass support structure **450** and the bonding interface **440**, and the bonding interface **440** may then be between the active layer **750** and the BEOL layer **430**. In some embodiments, some portions of the bonding interface **440** may be in contact with one or more portions of the active layer **750**, and other portions of the bonding interface **440** may be in contact with one or more portions of the BEOL layer **430**. The bonding interface **440** may be a hybrid bonding interface in such embodiments, e.g., as described below with reference to FIG. **10**.

As shown in FIG. **7**, the active layer **750** may include a plurality of interconnects **752**, arranged in one or more layers (two such layers are shown in FIG. **7**, separated by horizontal lines, but such separation may be absent in other embodiments, or other embodiments may include different number of layers and/or different arrangements of the interconnects **752** than what is shown in FIG. **7**). The interconnects **752** may include any suitable combination of vias **752-1** and lines **752-2**, some of which are labeled in FIG. **7** and some of which are not labeled in order to not clutter the drawing. The interconnects **752** may include any suitable conductive materials, such as any of the conductive metals or metal alloys as described above. Portions of various ones

of the interconnects **752** may be enclosed by an insulator material **754**, which may include any of the ILD materials described above. One or more of the interconnects **752** of the active layer **750** may be electrically coupled to (e.g., in electrically conductive contact with at least portions of) one or more of the plurality of BEOL interconnects **532**.

As further shown in FIG. 7, the IC assembly **700** may further include a plurality of devices **756**, such as transistors or memory cells. FIG. 7 illustrates the devices **756** as part of the glass support structure **450**, although, in other embodiments of the IC assembly **700**, the devices **756** may be a part of the active layer **750**. In some embodiments, one or more of the devices **756** may be transistors as described above, e.g., the transistors **100**. In some embodiments, one or more of the devices **756** may be memory cells as described above, e.g., the memory cells **300** or any other embedded memory cells. One or more of the interconnects **752** may then be coupled to one or more portions of the devices **756** and to one or more of the plurality of BEOL interconnects **532**.

In some embodiments, the cross-sectional side views of the interconnects **752** may have characteristic differences due to the fact that the interconnects **752** and the BEOL interconnects **532** are formed on different sides of the bonding interface **440**. In particular, in such embodiments, cross-sections, in a plane perpendicular to the FEOL layer **420**, of at least some of the interconnects **752** may be trapezoids with one short side and one long side. Such trapezoids may include two parallel sides, one of which is a short side and one of which is a longer side (i.e., the length of the longer side is greater than the length of the short side). Characteristic to the fact that interconnects **732** and the BEOL interconnects **532** are formed on different sides of the bonding interface **440**, for the trapezoids of the interconnects **732**, the short side would be closer to the glass support structure **450** than the long side and the long side would be closer to the bonding interface **440** and the FEOL layer **420** than the short side, whereas, for the trapezoids of the BEOL interconnects **532**, the long side would be closer to both the glass support structure **450** and the bonding interface **440** than the short side.

FIG. 8 provides a schematic illustration of an IC assembly **800** with back-side power delivery and glass support with thin-film devices and an active layer at the front, according to some embodiments of the present disclosure. Portions of the IC assembly **400** that were shown in FIG. 4, such as the back-side power delivery structure **410**, the FEOL **420**, etc., are labeled in the IC assembly **800** of FIG. 8. The IC assembly **800** further illustrates example implementations of each of these portions. In particular, the IC assembly **800** may be implemented as the IC assembly **600** with the one or more thin-film devices **556** disposed in the glass support structure **450** as described above, further including the active layer **750** of the IC assembly **700** as described above. This is shown in FIG. 8 with the IC assembly **800** including some of the same elements, showing using the same patterns, as those included in the IC assembly **600** illustrated in FIG. 6 and in the IC assembly **700** illustrated in FIG. 7. Descriptions of the IC assemblies with the one or more thin-film devices **556** disposed in the glass support structure **450** and with the active layer **750**, provided with reference to FIGS. 6 and 7, are applicable to the IC assembly **800** of FIG. 8 and, therefore, in the interests of brevity, are not repeated.

Example Fabrication Methods

IC assemblies with back-side power delivery and glass support at the front, as described herein, may be fabricated using any suitable techniques, e.g., subtractive, additive, damascene, dual damascene, etc. Some of such technique

may include suitable deposition and patterning techniques. As used herein, “patterning” may refer to forming a pattern in one or more materials using any suitable techniques (e.g., applying a resist, patterning the resist using lithography, and then etching the one or more material using dry etching, wet etching, or any appropriate technique).

FIGS. 9A-9D illustrate a first example method of forming an IC assembly with back-side power delivery and glass support at the front, according to some embodiments of the present disclosure. FIGS. 10A-10D illustrate a second example method of forming an IC assembly with back-side power delivery and glass support at the front, according to some embodiments of the present disclosure. The IC assemblies shown in FIGS. 9 and 10 include some of the same elements, showing using the same patterns, as those included in the IC assemblies shown in FIGS. 5-8. In the interests of brevity, detailed descriptions of those elements are applicable to the IC assemblies shown in FIGS. 9 and 10, and are not repeated.

FIG. 9A illustrates an IC structure **900A**, showing that the first fabrication method may begin with forming a plurality of FEOL devices **526** over a semiconductor support structure **902**, which may include any of the support structures described with reference to FIGS. 1-3, then forming the BEOL **430** over the FEOL layer **420** with the FEOL devices **526**, and then providing a layer of a bonding interface material **540** over the upper-most surface of the BEOL layer **430**. FIG. 9B illustrates an IC structure **900B**, showing that the first fabrication method may then proceed with flipping the IC structure **900A** of FIG. 9A upside down and bringing the bonding interface material **540** of the IC structure **900A** in contact with a bonding interface material provided at the upper surface of the glass support structure **450**, thus performing bonding of the IC structure **900A** and the glass support structure **450**. In general, bonding as described herein may be an insulator-insulator bonding, e.g., as oxide-oxide bonding, where a bonding interface material may be applied to one or both faces of the structures to be bonded, and then the structures are put together, possibly while applying a suitable pressure and heating up the assembly to a suitable temperature (e.g., to moderately high temperatures, e.g., between about 50 and 200 degrees Celsius) for a duration of time. In some embodiments, the bonding interface material **540** may be an adhesive material that ensures attachment of the IC structure **900A** and the glass support structure **450** to one another as shown in FIG. 9B and FIG. 9C. In some embodiments, the bonding interface material **540** may be an etch-stop material. In some embodiments, the bonding interface material **540** may be both an etch-stop material and have suitable adhesive properties to ensure attachment of the IC structures to one another as described herein. In some embodiments, no deliberately added adhesive bonding material may be used, in which case the layer labeled “**540**” or “**440**” in the present drawings represents a bonding interface resulting from the bonding of the respective IC structures to one another. Such a bonding interface may be recognizable as a seam or a thin layer in the IC assemblies described herein, using, e.g., selective area diffraction (SED), even when the specific materials of the insulators of the IC structures that are bonded together may be the same, in which case the bonding interface would still be noticeable as a seam or a thin layer in what otherwise appears as a bulk insulator (e.g., bulk oxide) layer. As used herein, unless specified otherwise, references to the “bonding interface material **540**” or the “bonding interface **440**” are applicable to a “bonding interface” for the embodiments where no deliberately added adhesive material is used to

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bond the IC structures as described herein. FIG. 9C illustrates an IC structure 900C, showing that, after bonding of the IC structure 900A and the glass support structure 450 has been performed, the first fabrication method may proceed with removing the semiconductor support structure 902 (e.g., using a suitable polishing or grinding process) to reveal back sides of the FEOL devices 526 of the FEOL 420. FIG. 9D illustrates an IC structure 900D, showing that, after the back sides of the FEOL devices 526 of the FEOL 420 have been revealed, the first fabrication method may proceed with providing the back-side power delivery structure 410 as described above.

FIG. 10A illustrates an IC structure 1000A, showing that the second fabrication method may begin with forming a plurality of FEOL devices 526 over a semiconductor support structure 902, which may include any of the support structures described with reference to FIGS. 1-3, and then forming the BEOL 430 over the FEOL layer 420 with the FEOL devices 526. FIG. 10B illustrates an IC structure 1000B, showing that the second fabrication method may then proceed with flipping the IC structure 1000A of FIG. 10A upside down and bringing the BEOL layer 430 of the IC structure 1000A in contact with the upper surface of the active layer 750 provided over the glass support structure 450, thus performing hybrid bonding of the IC structure 1000A and the glass support structure 450. Descriptions of bonding provided for FIGS. 9A-9D are applicable to the bonding of the IC structure 1000A and the glass support structure 450 and, therefore, in the interests of brevity, are not repeated. FIG. 10C illustrates an IC structure 1000C, showing that, after bonding of the BEOL layer 430 of the IC structure 900A and the active layer 750 has been performed, a bonding interface 440 may be formed between the active layer 750 and the BEOL layer 430. FIG. 10D illustrates that, subsequently, the second fabrication method may proceed with removing the semiconductor support structure 902 (e.g., using a suitable polishing or grinding process) to reveal back sides of the FEOL devices 526 of the FEOL 420 and, after that, providing the back-side power delivery structure 410 as described above.

Example Electronic Devices

IC assemblies with back-side power delivery and glass support at the front as disclosed herein may be included in any suitable electronic device. FIGS. 11-13 illustrate various examples of devices and components that may include one or more IC assemblies with back-side power delivery and glass support at the front as disclosed herein.

FIG. 11 is a side, cross-sectional view of an example IC package 2200 that may include one or more IC assemblies with back-side power delivery and glass support at the front in accordance with any of the embodiments disclosed herein. In some embodiments, the IC package 2200 may be a system-in-package (SiP).

The package substrate 2252 may be formed of a dielectric material (e.g., a ceramic, a buildup film, an epoxy film having filler particles therein, etc.), and may have conductive pathways extending through the dielectric material between the face 2272 and the face 2274, or between different locations on the face 2272, and/or between different locations on the face 2274.

The package substrate 2252 may include conductive contacts 2263 that are coupled to conductive pathways 2262 through the package substrate 2252, allowing circuitry within the dies 2256 and/or the interposer 2257 to electrically couple to various ones of the conductive contacts 2264 (or to other devices included in the package substrate 2252, not shown).

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The IC package 2200 may include an interposer 2257 coupled to the package substrate 2252 via conductive contacts 2261 of the interposer 2257, first-level interconnects 2265, and the conductive contacts 2263 of the package substrate 2252. The first-level interconnects 2265 illustrated in FIG. 13 are solder bumps, but any suitable first-level interconnects 2265 may be used. In some embodiments, no interposer 2257 may be included in the IC package 2200; instead, the dies 2256 may be coupled directly to the conductive contacts 2263 at the face 2272 by first-level interconnects 2265.

The IC package 2200 may include one or more dies 2256 coupled to the interposer 2257 via conductive contacts 2254 of the dies 2256, first-level interconnects 2258, and conductive contacts 2260 of the interposer 2257. The conductive contacts 2260 may be coupled to conductive pathways (not shown) through the interposer 2257, allowing circuitry within the dies 2256 to electrically couple to various ones of the conductive contacts 2261 (or to other devices included in the interposer 2257, not shown). The first-level interconnects 2258 illustrated in FIG. 11 are solder bumps, but any suitable first-level interconnects 2258 may be used. As used herein, a "conductive contact" may refer to a portion of electrically conductive material (e.g., metal) serving as an interface between different components; conductive contacts may be recessed in, flush with, or extending away from a surface of a component, and may take any suitable form (e.g., a conductive pad or socket).

In some embodiments, an underfill material 2266 may be disposed between the package substrate 2252 and the interposer 2257 around the first-level interconnects 2265, and a mold compound 2268 may be disposed around the dies 2256 and the interposer 2257 and in contact with the package substrate 2252. In some embodiments, the underfill material 2266 may be the same as the mold compound 2268. Example materials that may be used for the underfill material 2266 and the mold compound 2268 are epoxy mold materials, as suitable. Second-level interconnects 2270 may be coupled to the conductive contacts 2264. The second-level interconnects 2270 illustrated in FIG. 11 are solder balls (e.g., for a ball grid array arrangement), but any suitable second-level interconnects 2270 may be used (e.g., pins in a pin grid array arrangement or lands in a land grid array arrangement). The second-level interconnects 2270 may be used to couple the IC package 2200 to another component, such as a circuit board (e.g., a motherboard), an interposer, or another IC package, as known in the art and as discussed below with reference to FIG. 12.

The dies 2256 may take the form of any of the embodiments of the IC assemblies with back-side power delivery and glass support at the front discussed herein. In embodiments in which the IC package 2200 includes multiple dies 2256, the IC package 2200 may be referred to as a multi-chip package (MCP). The dies 2256 may include circuitry to perform any desired functionality. For example, one or more of the dies 2256 may be logic dies (e.g., silicon-based dies), and one or more of the dies 2256 may be memory dies (e.g., high bandwidth memory), including embedded logic and memory devices as described herein. In some embodiments, any of the dies 2256 may include one or more IC assemblies with back-side power delivery and glass support at the front, e.g., as discussed above; in some embodiments, at least some of the dies 2256 may not include any of the IC assemblies with back-side power delivery and glass support at the front.

The IC package 2200 illustrated in FIG. 11 may be a flip chip package, although other package architectures may be

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used. For example, the IC package **2200** may be a ball grid array (BGA) package, such as an embedded wafer-level ball grid array (eWLB) package. In another example, the IC package **2200** may be a wafer-level chip scale package (WLCS) or a panel fan-out (FO) package. Although two dies **2256** are illustrated in the IC package **2200** of FIG. **11**, an IC package **2200** may include any desired number of the dies **2256**. An IC package **2200** may include additional passive components, such as surface-mount resistors, capacitors, and inductors disposed on the first face **2272** or the second face **2274** of the package substrate **2252**, or on either face of the interposer **2257**. More generally, an IC package **2200** may include any other active or passive components known in the art.

FIG. **12** is a cross-sectional side view of an IC device assembly **2300** that may include components having one or more IC assemblies with back-side power delivery and glass support at the front in accordance with any of the embodiments disclosed herein. The IC device assembly **2300** includes a number of components disposed on a circuit board **2302** (which may be, e.g., a motherboard). The IC device assembly **2300** includes components disposed on a first face **2340** of the circuit board **2302** and an opposing second face **2342** of the circuit board **2302**; generally, components may be disposed on one or both faces **2340** and **2342**. In particular, any suitable ones of the components of the IC device assembly **2300** may include any of one or more IC assemblies with back-side power delivery and glass support at the front in accordance with any of the embodiments disclosed herein; e.g., any of the IC packages discussed below with reference to the IC device assembly **2300** may take the form of any of the embodiments of the IC package **2200** discussed above with reference to FIG. **11** (e.g., may include one or more IC assemblies with back-side power delivery and glass support at the front provided on a die **2256**).

In some embodiments, the circuit board **2302** may be a PCB including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **2302**. In other embodiments, the circuit board **2302** may be a non-PCB substrate.

The IC device assembly **2300** illustrated in FIG. **12** includes a package-on-interposer structure **2336** coupled to the first face **2340** of the circuit board **2302** by coupling components **2316**. The coupling components **2316** may electrically and mechanically couple the package-on-interposer structure **2336** to the circuit board **2302**, and may include solder balls (e.g., as shown in FIG. **12**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

The package-on-interposer structure **2336** may include an IC package **2320** coupled to an interposer **2304** by coupling components **2318**. The coupling components **2318** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **2316**. The IC package **2320** include one or more IC assemblies with back-side power delivery and glass support at the front as described herein. Although a single IC package **2320** is shown in FIG. **12**, multiple IC packages may be coupled to the interposer **2304**; indeed, additional interposers may be coupled to the interposer **2304**. The interposer **2304** may provide an intervening substrate used to bridge the circuit

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board **2302** and the IC package **2320**. Generally, the interposer **2304** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the interposer **2304** may couple the IC package **2320** (e.g., a die) to a BGA of the coupling components **2316** for coupling to the circuit board **2302**. In the embodiment illustrated in FIG. **12**, the IC package **2320** and the circuit board **2302** are attached to opposing sides of the interposer **2304**; in other embodiments, the IC package **2320** and the circuit board **2302** may be attached to a same side of the interposer **2304**. In some embodiments, three or more components may be interconnected by way of the interposer **2304**.

The interposer **2304** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, a ceramic material, or a polymer material such as polyimide. In some implementations, the interposer **2304** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **2304** may include metal interconnects **2308** and vias **2310**, including but not limited to through-silicon vias (TSVs) **2306**. The interposer **2304** may further include embedded devices **2314**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) protection devices, and memory devices. More complex devices such as radio frequency (RF) devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **2304**. The package-on-interposer structure **2336** may take the form of any of the package-on-interposer structures known in the art.

The IC device assembly **2300** may include an IC package **2324** coupled to the first face **2340** of the circuit board **2302** by coupling components **2322**. The coupling components **2322** may take the form of any of the embodiments discussed above with reference to the coupling components **2316**, and the IC package **2324** may take the form of any of the embodiments discussed above with reference to the IC package **2320**.

The IC device assembly **2300** illustrated in FIG. **12** includes a package-on-package structure **2334** coupled to the second face **2342** of the circuit board **2302** by coupling components **2328**. The package-on-package structure **2334** may include an IC package **2326** and an IC package **2332** coupled together by coupling components **2330** such that the IC package **2326** is disposed between the circuit board **2302** and the IC package **2332**. The coupling components **2328** and **2330** may take the form of any of the embodiments of the coupling components **2316** discussed above, and the IC packages **2326** and **2332** may take the form of any of the embodiments of the IC package **2320** discussed above. The package-on-package structure **2334** may be configured in accordance with any of the package-on-package structures known in the art.

FIG. **13** is a block diagram of an example computing device **2400** that may include one or more components with one or more IC assemblies with back-side power delivery and glass support at the front in accordance with any of the embodiments disclosed herein. Any of the components of the computing device **2400** may include an IC package **2200** as described with reference to FIG. **11**. Any of the compo-

nents of the computing device **2400** may include an IC device assembly **2300** as described with reference to FIG. 12.

A number of components are illustrated in FIG. 13 as included in the computing device **2400**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the computing device **2400** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single SoC die.

Additionally, in various embodiments, the computing device **2400** may not include one or more of the components illustrated in FIG. 13, but the computing device **2400** may include interface circuitry for coupling to the one or more components. For example, the computing device **2400** may not include a display device **2406**, but may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **2406** may be coupled. In another set of examples, the computing device **2400** may not include an audio input device **2418** or an audio output device **2408**, but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **2418** or audio output device **2408** may be coupled.

The computing device **2400** may include a processing device **2402** (e.g., one or more processing devices). As used herein, the term “processing device” or “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processing device **2402** may include one or more digital signal processors (DSPs), application-specific ICs (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The computing device **2400** may include a memory **2404**, which may itself include one or more memory devices such as volatile memory (e.g., DRAM), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some embodiments, the memory **2404** may include memory that shares a die with the processing device **2402**. This memory may be used as cache memory and may include one or more IC assemblies with back-side power delivery and glass support at the front as described herein.

In some embodiments, the computing device **2400** may include a communication chip **2412** (e.g., one or more communication chips). For example, the communication chip **2412** may be configured for managing wireless communications for the transfer of data to and from the computing device **2400**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

The communication chip **2412** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultramobile broadband (UMB) project (also referred

to as “3GPP2”), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **2412** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **2412** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **2412** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **2412** may operate in accordance with other wireless protocols in other embodiments. The computing device **2400** may include an antenna **2422** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

In some embodiments, the communication chip **2412** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **2412** may include multiple communication chips. For instance, a first communication chip **2412** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **2412** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication chip **2412** may be dedicated to wireless communications, and a second communication chip **2412** may be dedicated to wired communications.

The computing device **2400** may include battery/power circuitry **2414**. The battery/power circuitry **2414** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the computing device **2400** to an energy source separate from the computing device **2400** (e.g., AC line power).

The computing device **2400** may include a display device **2406** (or corresponding interface circuitry, as discussed above). The display device **2406** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display, for example.

The computing device **2400** may include an audio output device **2408** (or corresponding interface circuitry, as discussed above). The audio output device **2408** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds, for example.

The computing device **2400** may include an audio input device **2418** (or corresponding interface circuitry, as discussed above). The audio input device **2418** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

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The computing device **2400** may include a GPS device **2416** (or corresponding interface circuitry, as discussed above). The GPS device **2416** may be in communication with a satellite-based system and may receive a location of the computing device **2400**, as known in the art.

The computing device **2400** may include an other output device **2410** (or corresponding interface circuitry, as discussed above). Examples of the other output device **2410** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

The computing device **2400** may include an other input device **2420** (or corresponding interface circuitry, as discussed above). Examples of the other input device **2420** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

The computing device **2400** may have any desired form factor, such as a handheld or mobile computing device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a net-book computer, an ultrabook computer, a personal digital assistant (PDA), an ultramobile personal computer, etc.), a desktop computing device, a server or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable computing device. In some embodiments, the computing device **2400** may be any other electronic device that processes data.

SELECT EXAMPLES

The following paragraphs provide various examples of the embodiments disclosed herein.

Example 1 provides an IC assembly that includes an FEOL layer with a plurality of FEOL devices, a back-side power delivery structure with a plurality of power interconnects electrically coupled to (e.g., in electrically conductive contact with at least portions of) various ones of the plurality of FEOL devices, a BEOL layer with a plurality of BEOL interconnects electrically coupled to (e.g., in electrically conductive contact with at least portions of) one or more of the plurality of FEOL devices, and a glass support structure (e.g., at least a portion of a glass wafer), where the FEOL layer is between the back-side power delivery structure and the BEOL layer, and the BEOL layer is between the FEOL layer and the glass support structure.

Example 2 provides the IC assembly according to example 1, where the plurality of BEOL interconnects includes a first BEOL interconnect and a second BEOL interconnect (e.g., a first and a second metal lines), and the glass support structure includes a two-terminal thin-film device having a first terminal electrically coupled (e.g., in electrically conductive contact with) to the first BEOL interconnect and having a second terminal electrically coupled (e.g., in electrically conductive contact with) to the second BEOL interconnect.

Example 3 provides the IC assembly according to example 2, where the thin-film device is a thin-film resistor.

Example 4 provides the IC assembly according to example 2, where the thin-film device is a thin-film capacitor.

Example 5 provides the IC assembly according to example 2, where the thin-film device is a thin-film inductor.

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Example 6 provides the IC assembly according to any one of the preceding examples, further including a bonding interface between the BEOL layer and the glass support structure.

Example 7 provides the IC assembly according to example 6, where the bonding interface includes an oxide.

Example 8 provides the IC assembly according to example 7, where the oxide includes portions in contact with one or more portions of the glass support structure, and portions in contact with one or more portions of the BEOL layer.

Example 9 provides the IC assembly according to any one of examples 1-7, further including an active layer that includes a plurality of IC devices and interconnects, where the active layer is between the glass support structure and the bonding interface, the bonding interface is between the active layer and the BEOL layer, and at least one of the plurality of IC devices and interconnects of the active layer is electrically coupled to (e.g., in electrically conductive contact with at least portions of) one or more of the plurality of BEOL interconnects.

Example 10 provides the IC assembly according to example 9, where the bonding interface is a hybrid bonding interface.

Example 11 provides the IC assembly according to examples 9 or 10, where the bonding interface includes portions in contact with one or more portions of the active layer, and portions in contact with one or more portions of the BEOL layer.

Example 12 provides the IC assembly according to any one of examples 9-11, where a cross-section of each of at least one interconnect of the active layer and at least one interconnect of the BEOL interconnects is a trapezoid that includes two parallel sides, one of which is a short side and another one of which is a long side, for the trapezoid of the at least one interconnect of the active layer, the short side is closer to the glass support structure than the long side, and for the trapezoid of the at least one interconnect of the BEOL interconnects, the long side is closer to the glass support structure than the short side.

Example 13 provides the IC assembly according to example 12, where a cross-section of at least one interconnect of the power interconnects is a trapezoid that includes two parallel sides, one of which is a short side and another one of which is a long side, and, for the trapezoid of the at least one interconnect of the power interconnects, the short side is closer to the glass support structure than the long side.

Example 14 provides the IC assembly according to any one of the preceding examples, where the plurality of FEOL devices includes a FEOL transistor having a source region and a drain region, and at least one power interconnect of the plurality of power interconnects is electrically coupled to (e.g., in electrically conductive contact with) the source region or the drain region.

Example 15 provides the IC assembly according to any one of the preceding examples, where the back-side power delivery structure includes an insulator material enclosing at least portions of the plurality of power interconnects.

Example 16 provides the IC assembly according to any one of the preceding examples, where the BEOL layer includes one or more memory layers, the one or more memory layers including memory cells including thin-film transistors.

Example 17 provides the IC assembly according to any one of the preceding examples, where the glass support structure is replaced with a support structure of a material having a dielectric constant lower than 10, which may, but

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does not have to be, glass. For example, the material of the support structure may be mica.

Example 18 provides an IC package that includes an IC assembly according to any one of the preceding examples; and a further IC component, coupled to the IC assembly.

Example 19 provides the IC package according to example 18, where the further IC component includes one of a package substrate, an interposer, or a further IC die.

Example 20 provides the IC package according to examples 18 or 20, where the IC assembly includes, or is a part of, at least one of a memory device, a computing device, a wearable device, a handheld electronic device, and a wireless communications device.

Example 21 provides an electronic device that includes a carrier substrate; and one or more of the IC assembly according to any one of the preceding examples and the IC package according to any one of the preceding examples, coupled to the carrier substrate.

Example 22 provides the electronic device according to example 21, where the carrier substrate is a motherboard.

Example 23 provides the electronic device according to example 21, where the carrier substrate is a PCB.

Example 24 provides the electronic device according to any one of examples 21-23, where the electronic device is a wearable electronic device (e.g., a smart watch) or handheld electronic device (e.g., a mobile phone).

Example 25 provides the electronic device according to any one of examples 21-24, where the electronic device further includes one or more communication chips and an antenna.

Example 26 provides the electronic device according to any one of examples 21-25, where the electronic device is an RF transceiver.

Example 27 provides the electronic device according to any one of examples 21-25, where the electronic device is one of a switch, a power amplifier, a low-noise amplifier, a filter, a filter bank, a duplexer, an upconverter, or a down-converter of an RF communications device, e.g. of an RF transceiver.

Example 28 provides the electronic device according to any one of examples 21-25, where the electronic device is a computing device.

Example 29 provides the electronic device according to any one of examples 21-28, where the electronic device is included in a base station of a wireless communication system.

Example 30 provides the electronic device according to any one of examples 21-28, where the electronic device is included in a user equipment device (i.e., a mobile device) of a wireless communication system.

Example 31 provides a method of fabricating an IC assembly, the method including: providing FEOL devices over a semiconductor support structure; providing a BEOL layer over the FEOL devices, the BEOL layer including a plurality of BEOL interconnects electrically coupled to (e.g., in electrically conductive contact with at least portions of) one or more of the plurality of FEOL devices; bonding an arrangement of the BEOL layer and the FEOL devices to a non-semiconductor support structure; performing a back-side reveal by removing at least portions of the semiconductor support structure to expose portions of the FEOL devices; and providing a back-side power delivery structure, including a plurality of power interconnects electrically coupled to (e.g., in electrically conductive contact with at least portions of) the exposed portions of the FEOL devices.

Example 32 provides the method according to example 31, where bonding the arrangement of the BEOL layer and

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the FEOL devices to the non-semiconductor support structure includes providing one or more bonding materials on at least one of a face of the BEOL layer to be bonded to the non-semiconductor support structure and a face of the non-semiconductor support structure to be bonded to the BEOL layer, and attaching the face of the BEOL layer to be bonded to the non-semiconductor support structure to the face of the non-semiconductor support structure to be bonded to the BEOL layer.

Example 33 provides the method according to example 32, where the one or more bonding materials include an oxide.

Example 34 provides the method according to any one of examples 31-33, where removing the at least portions of the semiconductor support structure to expose the portions of the FEOL devices includes polishing or grinding away the semiconductor support structure until the portions of the FEOL devices are exposed.

Example 35 provides the method according to any one of examples 31-34, where the at least portions of the semiconductor support structure are removed after bonding the arrangement of the BEOL layer and the FEOL devices to the non-semiconductor support structure.

Example 36 provides the method according to any one of examples 31-35, where the back-side power delivery structure includes an insulator material enclosing at least portions of the plurality of power interconnects.

Example 37 provides the method according to any one of examples 31-36, where the non-semiconductor support structure includes glass.

Example 38 provides the method according to any one of examples 31-37, where the non-semiconductor support structure includes mica.

Example 39 provides the method according to any one of examples 31-37, where the non-semiconductor support structure includes an active layer that includes a plurality of IC devices and interconnects, and bonding the arrangement of the BEOL layer and the FEOL devices to the non-semiconductor support structure includes bonding the arrangement of the BEOL layer and the FEOL devices to the active layer to electrically couple at least one of the plurality of IC devices and interconnects of the active layer and one or more of the plurality of BEOL interconnects.

Example 40 provides the method according to any one of examples 31-39, further including processes for forming the IC assembly according to any one of the preceding examples (e.g., for forming the IC assembly according to any one of examples 1-17).

The above description of illustrated implementations of the disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. While specific implementations of, and examples for, the disclosure are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize. These modifications may be made to the disclosure in light of the above detailed description.

The invention claimed is:

1. An integrated circuit (IC) assembly, comprising:
 - a front end of line (FEOL) layer, comprising a plurality of FEOL devices;
 - a back-side power delivery structure, comprising a plurality of power interconnects coupled to various of the plurality of FEOL devices;
 - a back end of line (BEOL) layer, comprising a plurality of BEOL interconnects coupled to one or more of the

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plurality of FEOL devices, wherein the plurality of BEOL interconnects includes a first BEOL interconnect and a second BEOL interconnect;

a glass support structure comprising a thin-film device having a first terminal coupled to the first BEOL interconnect and having a second terminal coupled to the second BEOL interconnect; and

a bonding interface between the BEOL layer and the glass support structure,

wherein:

- the FEOL layer is between the back-side power delivery structure and the BEOL layer,
- the BEOL layer is between the FEOL layer and the glass support structure, and
- at least a portion of the thin-film device extends through the bonding interface.

2. The IC assembly according to claim 1, wherein the thin-film device is one of a resistor, a capacitor, or an inductor.

3. The IC assembly according to claim 1, wherein the bonding interface includes oxygen.

4. The IC assembly according to claim 1, wherein:

- the plurality of FEOL devices includes a FEOL transistor having a source region and a drain region, and
- at least one power interconnect of the plurality of power interconnects is coupled to the source region or the drain region.

5. The IC assembly according to claim 1, wherein the back-side power delivery structure includes an insulator material enclosing at least portions of the plurality of power interconnects.

6. The IC assembly according to claim 1, wherein the BEOL layer includes one or more memory layers, the one or more memory layers including memory cells comprising thin-film transistors.

7. The IC assembly according to claim 1, further comprising:

- a conductive interconnect extending through the bonding interface,

wherein the first terminal of the thin-film device is coupled to the first BEOL interconnect via the conductive interconnect.

8. The IC assembly according to claim 1, wherein the bonding interface is a hybrid bonding interface.

9. The IC assembly according to claim 1, further comprising a layer that includes a plurality of IC devices and interconnects, wherein:

- the layer is between the glass support structure and the bonding interface,
- the bonding interface is between the layer and the BEOL layer, and
- at least one of the plurality of IC devices and interconnects of the layer is coupled to one or more of the plurality of BEOL interconnects.

10. The IC assembly according to claim 9, wherein the bonding interface includes:

- portions in contact with one or more portions of the layer, and
- portions in contact with one or more portions of the BEOL layer.

11. The IC assembly according to claim 9, wherein:

- a cross-section of each of at least one interconnect of the layer and at least one interconnect of the BEOL interconnects is substantially a trapezoid that includes two parallel sides, one of which is a short side and another one of which is a long side,

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for the trapezoid of the at least one interconnect of the layer, the short side is closer to the glass support structure than the long side, and

for the trapezoid of the at least one interconnect of the BEOL interconnects, the long side is closer to the glass support structure than the short side.

12. The IC assembly according to claim 11, wherein:

- a cross-section of at least one interconnect of the power interconnects is a trapezoid that includes two parallel sides, one of which is a short side and another one of which is a long side, and
- for the trapezoid of the at least one interconnect of the power interconnects, the short side is closer to the glass support structure than the long side.

13. An integrated circuit (IC) assembly, comprising:

- a first layer, comprising a plurality of transistors;
- a power delivery structure, comprising a plurality of power interconnects coupled to at least one of the transistors;
- a second layer, comprising a plurality of conductive interconnects coupled to one or more of the plurality of transistors, wherein the first layer is between the power delivery structure and the second layer;
- a glass structure comprising a device having a terminal coupled to at least one of the plurality of interconnects; and
- a bonding interface between the second layer and the glass structure,

wherein:

- the at least one of the plurality of interconnects extends through the bonding interface, or
- the terminal of the device is coupled to the at least one of the plurality of interconnects via a conductive interconnect that extends through the bonding interface.

14. The IC assembly according to claim 13, wherein the device is one of a transistor, a memory cell, a resistor, a capacitor, an inductor.

15. The IC assembly according to claim 13, wherein the bonding interface includes oxygen.

16. An integrated circuit (IC) assembly, comprising:

- a first layer comprising a plurality of devices, wherein the plurality of devices include transistors or memory cells;
- an interconnect structure comprising a plurality of conductive interconnects coupled to one or more of the plurality of devices;
- a second layer comprising a first conductive interconnect and a second conductive interconnect;
- a glass layer comprising a device having a first terminal coupled to the first conductive interconnect and having a second terminal coupled to the second conductive interconnect;
- a bonding interface between the second layer and the glass layer; and
- a third conductive interconnect extending through the bonding interface,

wherein:

- the first layer is between the interconnect structure and the second layer,
- the second layer is between the first layer and the glass layer, and
- the first terminal of the device is coupled to the first conductive interconnect via the third conductive interconnect.

17. The IC assembly according to claim 16, wherein at least a portion of the device extends through the bonding interface.

18. The IC assembly according to claim **16**, wherein the device is one of a transistor, a memory cell, a resistor, a capacitor, an inductor.

19. The IC assembly according to claim **16**, wherein the bonding interface is a hybrid bonding interface. 5

20. The IC assembly according to claim **16**, wherein the bonding interface includes oxygen.

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